How fiber orientation steers in-plane elastic waves: energy channeling, its limits, and tow-curvature-constrained designs

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Abstract

Question. How much control over an in-plane elastic wave field can the fiber orientation of a continuous-fiber composite exert, at fixed material and mass and under the tow-curvature limit of automated placement? By what mechanism, and within what limits?
Hypothesis. In a strongly anisotropic lamina the energy of a quasi-longitudinal (qP) wave is skewed toward the fibers, by up to 65° for CFRP, so that 88% of wave normals carry their energy within 20° of them. Where the field is qP-dominated and the fibers turn slowly on the wavelength scale, a fiber field should act as a designable field of energy streamlines, and its robust effect should be broadband *redirection* of energy.
Analysis. We solve the two-dimensional orthotropic time-harmonic problem in `dolfinx` with discrete adjoints, a smooth orientation field and a pointwise tow-curvature constraint. Every focus gain is split exactly into redirection (the target's share of the energy) and whole-field growth, and checked against the share of injected power absorbed in the target, a frequency sweep, the energy-flux field, a ray prediction from the fiber field alone, a series of absorbing layers, anisotropy and curvature studies with multiple starts, and an as-built tow model.
Results. With an absorbing layer whose reflection is converged ($|R| \leq 0.04$), an orientation-only lens redirects 3.9–4.0× more energy into its target, band-averaged over the tested wavelengths, and absorbs 3.2–12.8× more of the injected power there. Rays traced through the fiber field predict 4.2–4.7×. The energy flux stays within about 5.8° of the fibers wherever they turn by less than about 4 rad per wavelength. The 78× single-frequency focus gain is mostly a resonance of the finite plate (quality factor ≈ 26). Redirection vanishes for isotropic material, grows monotonically with anisotropy (best of four starts), and is unaffected while the minimum tow radius is below about 0.7 wavelengths. A full cloak reaches 36× far-field reduction only at its design frequency and on its design mesh (24× on a mesh 1.8 times finer).
Conclusion. Fiber orientation steers in-plane elastic energy by channeling it along the fibers, a few-fold and broadband in the ray regime. Large single-frequency gains of finite plates are mostly resonance and should not be reported as steering.

Keywords: elastic waves; fiber steering; anisotropic plates; energy channeling; topology optimization; cloaking; automated fiber placement.

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1 Introduction

Motivation. Automated fiber placement and continuous-fiber printing lay composite tows along curved paths, so the local fiber orientation $\theta(\boldsymbol{x})$ of a part becomes a design field. That field is routinely designed for *static* performance: stiffness and strength under a minimum turning-radius constraint, with the toolpath carried through the optimization [11]. A carbon/epoxy lamina is also strongly *acoustically* anisotropic. Its quasi-longitudinal wave speed differs by a factor of almost four between the fiber and transverse directions. This raises the possibility that the same manufacturing freedom, with no change of material or mass, could steer in-plane elastic waves: concentrate vibration where it is wanted, keep it from where it is not, or hide an obstacle from it.

1.1 Related work and positioning

Fiber-orientation design. Curvilinear fiber formats were introduced to raise buckling loads and relieve stress concentrations [1]. Variable-stiffness laminates are now designed routinely for stiffness and buckling [2]. Orientation is also optimized element-wise in discrete-material and continuous-orientation topology optimization [3, 4]. Continuous-fiber additive manufacturing [5] has made arbitrary in-plane fiber fields buildable. Wong et al. integrated toolpath generation and a local fiber-curvature limit into compliance-minimizing topology optimization [11]. We adopt their CS-RBF orientation field and curvature constraint. Smoothness and manufacturability of optimized orientation fields are active topics in their own right: smooth-orientation filters in anisotropic topology optimization [13], segmentation of an optimal stiffness field into manufacturable curvilinear patches [14], and models of the gaps and overlaps of tow placement [16, 15]. Curvature-constrained orientation design is therefore not new, and we do not claim it. This work, like most of the literature, targets *static*, *modal* or forced-vibration objectives: the fundamental frequency [6, 15], the natural modes of curvilinear-fiber laminates [7], and minimum sound radiation from vibrating laminates by discrete orientation selection [8]. Anisotropic composite layouts have also been designed to shield, focus and cloak *diffusive* heat flux [9, 10]. Lu and Tong optimized macrostructure, microstructure and spatially varying orientation of anisotropic material concurrently for mechanical cloaking [12], and Olsson and Wall chose fiber directions for exact partial cloaking in the inextensible limit [21]. Our problem differs from these in three ways: the field is time-harmonic in-plane elastodynamics, the orientation of one fixed lamina is the *only* design field (no density, topology or material change), and the field must respect the tow-curvature limit. A targeted search found no prior work that optimizes fiber orientation alone under that limit to steer elastic waves; the search was not exhaustive.

Wave control by graded media. Transformation optics [17] and transformation elastodynamics [18, 19, 26, 20] prescribe material fields that bend waves arbitrarily. The required media are singular at the inner surface of a cloak. With isotropic stiffness the density must become infinite there, and a finite-mass cloak needs anisotropic (pentamode) stiffness instead [19]; for in-plane elasticity the transform keeps the Navier form but with a tensor that is singular at the inner boundary and lacks the minor symmetries of a Cauchy solid [26, 20], and in general the transformed equations are not of Cauchy-elastic form [18], so no lamina can realize them. Pre-stressing a hyperelastic solid is another route to the required properties [24, 25]. Olsson and Wall showed that inextensible fibers in a variable direction give form-invariant equations of motion and exact partial cloaking at all frequencies [21]; this is the idealized limit of unbounded anisotropy, which a real lamina does not reach. Ground-plane (carpet) cloaks, which only need to mimic a flat reflecting surface, can be broadband with bounded (natural or layered) anisotropy [22, 23]. In practice, elastic lenses and wave guides use graded index fields made from thickness variations [27] or phononic crystals [28], mostly for flexural waves in isotropic plates. Adjoint inverse design of plate wave devices has used pinning, mass loading and grading [29]. Topology optimization of wave problems is well developed for band gaps, acoustics and photonics [30, 31, 35], including low-contrast optical cloaks [32] and acoustic cloaks [33], and the single-frequency optima it produces are known to be narrowband, which motivates band-averaged objectives [34, 33]. In these approaches the design variable is the *amount* of material or its local stiffness, not its *orientation*.

Waves in anisotropic plates. In anisotropic solids, energy does not travel along the wave normal: the group velocity is skewed toward stiff directions [36, 37]. In fiber-reinforced composite plates this focuses guided waves into the directions of minimum slowness, which are the stiff directions [38]. In some anisotropic plates the power flux of slow guided waves can even be perpendicular to the wave vector [39]. In non-destructive evaluation the skew is a nuisance to be compensated. Here it is the mechanism to be designed.

What is new here. The ingredients are known: group-velocity skew in anisotropic solids [36, 38], inverse design of wave devices [30, 29], and curvature-constrained orientation fields [11]. The contribution is their combination for a *wave-control target* with orientation as the only design field, and an analysis of *energy transport* that separates what the fibers do from what the finite plate does. Specifically: (i) the mechanism, energy channeling by group-velocity skew, is tested prediction by prediction, including against a ray prediction computed from the fiber field alone and against fields that were never optimized; (ii) an exact identity splits every focus gain into redirection and resonant whole-field growth, and the redirection is cross-checked against the share of injected power the target absorbs; (iii) following band-objective practice [34, 33], every design is evaluated across a band and against a series of absorbing layers. The orientation fields satisfy the tow-curvature limit pointwise; we call them *tow-curvature-constrained* rather than manufacturable, because tow width, coverage and defects are only estimated (Sec. 6.6).

1.2 Question, hypothesis and predictions

Scientific question.

> *How much control over an in-plane elastic wave field can the fiber orientation of a continuous-fiber composite exert, at fixed material and mass and under the tow-curvature limit of automated placement? By what mechanism does that control act, and what bounds it?*

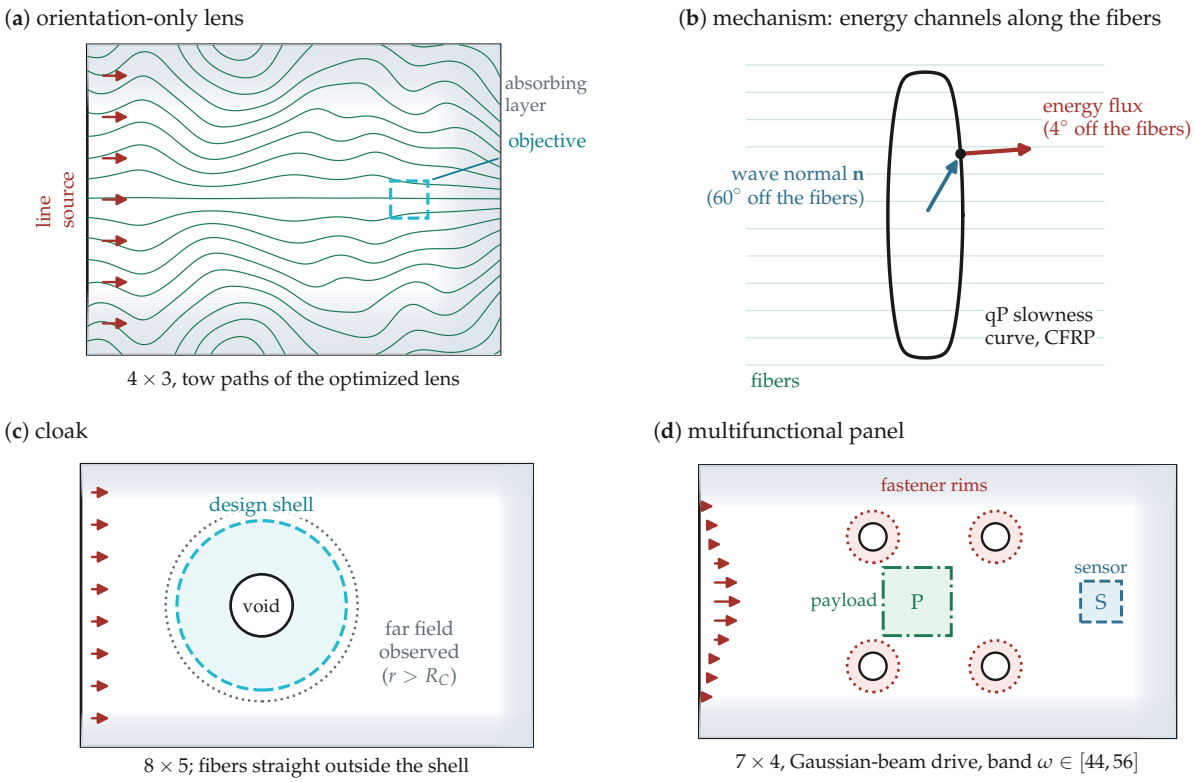


Figure 1: The three problems and the mechanism. **(a)** The lens: a 4×3 CFRP plate driven by an x-directed line source at its left edge, with absorbing layers on the other three edges (shaded), the tow paths of the optimized lens, and the objective region (dashed). **(b)** Why fibers channel energy: the quasi-P slowness curve of the CFRP lamina (fibers along x). The energy of a plane wave travels along the outward normal to this curve, so a wave whose normal is 60° off the fibers carries its energy only a few degrees off them (Sec. 3). **(c)** The cloak: a traction-free void cut from the mesh, the design shell in which the orientation may change (dashed), and the far-field observation region beyond R_C (dotted). **(d)** The multifunctional panel: four fastener holes with their rim regions (dotted), the payload P and sensor S, and a Gaussian-beam drive.

Hypothesis. (Fig. 1.) In a strongly anisotropic lamina, the energy of a plane wave does not travel along its wave normal. It is skewed toward the fiber direction, by up to 65° for CFRP (Sec. 3). Most wave normals therefore carry their energy within a narrow cone about the fibers. We hypothesize that a fiber field acts as a field of *energy streamlines*: designing $\theta(\boldsymbol{x})$ designs where vibration energy goes, because energy is channeled along the fibers. The hypothesis has stated conditions. It concerns the qP part of the field. It is a ray (WKB) statement, so it requires the fibers to turn slowly on the scale of the wavelength. And it ignores reflections and interference. Where these conditions fail (long wavelengths, sharply turning fibers, standing-wave nodes, boundaries) we expect channeling to fail, and we measure where it does. Section 3 makes this precise. It predicts:

W1 An isotropic plate offers no orientation control at all.

W2 The robust effect of an optimized field is *redirection*: a larger share of the wave energy, and of the injected power, in the target. Because the energy skew of a homogeneous lamina does

not depend on frequency, redirection should be broadband over the wavelengths for which the ray picture holds. Any gain beyond redirection that appears at the design frequency is whole-field build-up, consistent with resonance, and narrowband.

W3 In the optimized designs, the local energy flux follows the fiber direction where the fibers turn slowly, and a ray prediction from the fiber field alone reproduces the redirection.

W4 Redirection grows with the lamina anisotropy, tracking the share of wave normals that the lamina channels toward its fibers.

W5 Energy can follow a curving fiber field only as fast as the field may turn, so the tow-curvature limit bounds the redirection. Canceling the field of a finite scatterer in free space (a full cloak, not a ground-plane cloak) needs the singular anisotropy of transformation media. With the bounded anisotropy of a real lamina it must rely on interference, and we predict it to be narrowband.

1.3 Approach and findings

Approach. We solve the genuine two-dimensional orthotropic time-harmonic problem: the rotated plane-stress stiffness $\boldsymbol{D}_f(\theta)$ with $E_{f1}/E_{f2} = 14.6$, light structural damping, and an absorbing sponge. A real four-field split runs it on a real-arithmetic `PETSc/dolfinx` build, and automatic discrete adjoints give the orientation sensitivity. The field is parametrized by smooth compactly supported radial basis functions and bounded by the tow curvature. We optimize a lens and a cloak, then test each prediction by decomposing the gains, sweeping frequency, measuring the energy-flux field, varying the anisotropy, and tightening the curvature limit, each with multiple optimizer starts where the result depends on them. A band-averaged multifunctional panel shows how the results translate into a design practice.

Organization. Section 2 states the model and its physical scales, Sec. 3 the channeling theory and the ray prediction, Secs. 4–5 the discretization, adjoint and orientation parametrization, Sec. 6 the tests of W1–W5, the as-built check and the multifunctional panel, and Sec. 7 what is and is not established.

2 Governing model and the anisotropy lever

Consider in-plane time-harmonic motion $\mathrm{Re}\{\mathbf{u}(\mathbf{x})e^{+\mathrm{i}\omega t}\}$ (with this sign convention, structural damping $(1+\mathrm{i}\eta)$ and a viscous sponge $+\mathrm{i}\omega c$ are both dissipative), $\mathbf{u} = (u_x, u_y)$, of a thin composite plate $\Omega \subset \mathbb{R}^2$ under plane-stress. With the engineering Voigt strain

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \left[\, \partial_x u_x,\ \ \partial_y u_y,\ \ \partial_y u_x + \partial_x u_y \,\right]^{\mathsf{T}}, \tag{1}$$

the fiber composite has the rotated orthotropic plane-stress stiffness

$$\mathbf{D}_f(\theta) = \mathbf{T}(\theta)\,\mathbf{Q}_f\,\mathbf{T}(\theta)^{\mathsf{T}}, \qquad \mathbf{T}(\theta) = \begin{bmatrix} c^2 & s^2 & -2sc \\ s^2 & c^2 & 2sc \\ sc & -sc & c^2 - s^2 \end{bmatrix}, \quad (c,s) = (\cos\theta, \sin\theta), \tag{2}$$

where $\mathbf{Q}_f$ is the fiber stiffness in material axes,

$$\mathbf{Q}_f = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix}, \quad Q_{11} = \tfrac{E_{f1}}{1-\nu_{12}\nu_{21}},\ \ Q_{22} = \tfrac{E_{f2}}{1-\nu_{12}\nu_{21}},\ \ Q_{12} = \tfrac{\nu_{12}E_{f2}}{1-\nu_{12}\nu_{21}},\ \ Q_{66} = G_{12}, \tag{3}$$

Table 1: Wavelengths and resolution of the three problems (lengths in units of L_0, frequency by (5)). $\lambda_\parallel$, $\lambda_\perp$: qP along and across the fibers; λ_{qS}: the shortest (quasi-shear) wavelength; h: element size; $L_\theta = 1/\max|\nabla\theta|$: the shortest orientation-gradient length of the design.

problem	domain	ω	$\lambda_\parallel$	$\lambda_\perp$	λ_{qS}	h	λ_{qS}/h	L_θ
lens	4×3	40	1.42	0.37	0.28	0.031	9	0.15
cloak	8×5	50	1.14	0.30	0.22	0.045	4.9	–
panel	7×4	50	1.14	0.30	0.22	0.055	4.0	0.17

with $\nu_{21} = \nu_{12}E_{f2}/E_{f1}$. We use a carbon/epoxy lamina $E_{f1} = 131$, $E_{f2} = 9$, $G_{12} = 5$, $\nu_{12} = 0.27$ (a modulus contrast $E_{f1}/E_{f2} \approx 14.6$), density $\rho = 1.6$. The time-harmonic displacement obeys

$$-\nabla\cdot\big[(1+\mathrm{i}\eta)\,\boldsymbol{\sigma}(\mathbf{u};\theta)\big] - \omega^2\rho\,\mathbf{u} + \mathrm{i}\,\omega\,c\,\mathbf{u} = \mathbf{f}, \qquad \boldsymbol{\sigma} = \mathbf{D}_f(\theta)\,\boldsymbol{\varepsilon}(\mathbf{u}), \tag{4}$$

where η is a small structural (hysteretic) damping that keeps the response off sharp undamped resonances (we use $\eta = 0.02$ in every computation, and test 0.01 and 0.04 in Appendix F; without it the field is speckle), and $c(\mathbf{x}) \geq 0$ is a mass-proportional absorbing sponge (§4.3). The orientation $\theta(\mathbf{x})$ is the sole design field.

Units and physical scales. The computations are nondimensional. Moduli are in GPa and density in $10^3\,\mathrm{kg/m^3}$, so speeds come out in km/s: the qP speed is 9.07 along the fibers and 2.38 across them, and the slowest quasi-shear speed is 1.77. Lengths are in an arbitrary unit L_0, so time is in $L_0/(1\,\mathrm{km/s})$ and a nondimensional angular frequency ω is the physical frequency

$$f = \frac{\omega}{2\pi}\,\frac{1\,\mathrm{km/s}}{L_0}. \tag{5}$$

Every design is defined by its wavelengths, and Table 1 lists them. The tow-curvature results of Sec. 6.4 are stated as a minimum tow radius R in along-fiber wavelengths, which is the quantity that transfers between scales. Fixing L_0 by the tow radius a process can achieve sets the plate size and frequency. Redirection survives down to $R \approx 0.7\lambda_\parallel = 1.0\,L_0$ (Sec. 6.4). Continuous-fiber printing with $R_{\min} \approx 20\,\mathrm{mm}$ therefore allows $L_0 \geq 20\,\mathrm{mm}$: a $80\times60\,\mathrm{mm}$ lens at $f \approx 320\,\mathrm{kHz}$. Automated fiber placement of 6.35 mm tows with $R_{\min} \approx 0.5\,\mathrm{m}$ requires $L_0 \geq 0.5\,\mathrm{m}$: a $2\times1.5\,\mathrm{m}$ panel at $f \approx 13\,\mathrm{kHz}$.

Remark 1 (Isotropy annihilates the orientation lever)**.** *The stiffness of an* isotropic *material is a rotationally invariant tensor: if $\mathbf{Q}_f$ is isotropic ($Q_{11} = Q_{22}$ and $Q_{66} = \frac{1}{2}(Q_{11} - Q_{12})$) then $\mathbf{T}(\theta)\mathbf{Q}_f\mathbf{T}(\theta)^\mathsf{T} = \mathbf{Q}_f$ for every θ, so $\mathbf{D}_f(\theta)$ — and hence the entire operator (4) — is* independent of θ. *Consequently $\partial\mathbf{D}_f/\partial\theta \equiv \mathbf{0}$, every orientation sensitivity below (which we show is built from $\partial\mathbf{D}_f/\partial\theta$) vanishes identically, and fiber orientation is a* dead design variable *for an isotropic plate. Fiber anisotropy is therefore* necessary *for orientation-based wave control. How much control it buys is the subject of Sec. 3 and of the anisotropy study of Sec. 6.3.*

3 Theory: how a fiber field steers wave energy

3.1 Plane waves in a rotated lamina

Consider a homogeneous lamina with fiber angle θ and a plane wave $\mathbf{u} = \boldsymbol{a}\,e^{\mathrm{i}(\omega t - k\,\mathbf{n}\cdot\mathbf{x})}$ with unit wave normal $\mathbf{n} = (\cos\phi, \sin\phi)$. The engineering strain is $\boldsymbol{\varepsilon} = -\mathrm{i}k\,\mathbf{N}(\mathbf{n})\boldsymbol{a}$, with $\mathbf{N}(\mathbf{n}) = \begin{bmatrix} n_1 & 0 \\ 0 & n_2 \\ n_2 & n_1 \end{bmatrix}$.

Substituting into (4) without damping gives the Christoffel problem

$$\boldsymbol{\Gamma}(\mathbf{n};\theta)\,\boldsymbol{a} = \rho c^2 \boldsymbol{a}, \qquad \boldsymbol{\Gamma}(\mathbf{n};\theta) = \mathbf{N}(\mathbf{n})^{\mathsf{T}}\mathbf{D}_f(\theta)\,\mathbf{N}(\mathbf{n}), \qquad c = \omega/k. \tag{6}$$

Its two eigenvalues give the phase speeds of the quasi-longitudinal (qP) and quasi-shear (qS) waves.

Proposition 1 (Only the relative angle matters)**.** *$\boldsymbol{\Gamma}(\mathbf{n};\theta)$ is unitarily similar to $\boldsymbol{\Gamma}(\mathbf{n}_\psi;0)$, where $\mathbf{n}_\psi$ is the unit vector at the relative angle $\psi = \phi - \theta$. Hence $c = c(\psi)$, and for an isotropic lamina c is independent of ψ.*

Proof. Let $\boldsymbol{R}$ be the in-plane rotation by θ. Then $\mathbf{N}(\boldsymbol{R}\mathbf{n}_\psi) = \mathbf{T}(\theta)^{-\mathsf{T}}\mathbf{N}(\mathbf{n}_\psi)\boldsymbol{R}^{\mathsf{T}}$ (strain rotation), and $\mathbf{D}_f(\theta) = \mathbf{T}\mathbf{Q}_f\mathbf{T}^{\mathsf{T}}$. Hence $\boldsymbol{\Gamma}(\mathbf{n};\theta) = \boldsymbol{R}\,\mathbf{N}(\mathbf{n}_\psi)^{\mathsf{T}}\mathbf{Q}_f\mathbf{N}(\mathbf{n}_\psi)\boldsymbol{R}^{\mathsf{T}} = \boldsymbol{R}\,\boldsymbol{\Gamma}(\mathbf{n}_\psi;0)\,\boldsymbol{R}^{\mathsf{T}}$. If $\mathbf{Q}_f$ is isotropic, then $\mathbf{D}_f(\theta) = \mathbf{Q}_f$ (Remark 1) and $\boldsymbol{\Gamma}$ depends on $\mathbf{n}$ only through its rotation. □

3.2 Energy velocity and skew

For a lossless homogeneous medium, the time-averaged energy flux of a plane wave equals its mean energy density times the group velocity $\boldsymbol{c}_g = \nabla_{\boldsymbol{k}}\omega$ [36]. Writing $\omega = k\,c(\psi)$ in polar wave-vector coordinates,

$$\boldsymbol{c}_g = c(\psi)\,\mathbf{n} + c'(\psi)\,\mathbf{n}^{\perp}, \qquad \tan\delta(\psi) = \frac{c'(\psi)}{c(\psi)}, \tag{7}$$

so the energy travels at the angle $\psi + \delta(\psi)$ from the fiber. Here δ is the *skew* between the energy direction and the wave normal. For the CFRP lamina ($E_{f1}/E_{f2} = 14.6$) the qP speed falls from 9.07 along the fiber to 2.38 across it, and the skew reaches 65° at $\psi = 75°$.

- A qP wave whose normal is 60° off the fibers carries its energy only 4° off them.
- The qS skew reaches 59°.

Channeling. We call the fraction of wave normals whose energy direction lies within 20° of the fiber the *channeling share* s_{20} of the lamina. For an isotropic lamina it is 22%, the purely geometric fraction 20/90. Along the family $\mathbf{Q}(s)$, which keeps the isotropic invariants and scales the anisotropic ones (Sec. 6.3), it rises monotonically:

E_{f1}/E_{f2}	1.0	1.7	2.9	5.4	14.6	31.9
max qP skew	0°	18°	35°	50°	65°	71°
s_{20}	22%	42%	60%	75%	88%	94%

A CFRP lamina therefore sends the energy of almost every incident direction along its fibers. This is the physical content of the hypothesis.

Rays in a steered lamina, and when they apply. When $\theta(\mathbf{x})$ varies slowly on the wavelength scale, ray (WKB) theory applies locally. qP energy then travels along the rays of the Hamiltonian $\omega(\mathbf{x},\boldsymbol{k}) = |\boldsymbol{k}|\,c(\psi - \theta(\mathbf{x}))$, with ψ the wave-normal angle and c the qP phase speed of the lamina. Since ω is conserved, Hamilton's equations reduce to

$$\dot{\mathbf{x}} = c\,\mathbf{n} + c'\,\mathbf{n}^{\perp}, \qquad \dot{\psi} = c'(\psi - \theta)\,\big(\mathbf{n}^{\perp}\cdot\nabla\theta\big), \tag{8}$$

the first being the group velocity (7). Because c' is small except near $\psi - \theta \approx 75°$, rays are dragged along with the fibers, and the energy paths approximate the integral curves of the fiber field.

Launching rays from the drive and absorbing them in the sponge, the target's share of the ray dwell time is a single-pass, frequency-independent estimate of its share of the energy. Its ratio to straight fibers is a ray prediction G_{ray} of the redirection that uses only the fiber field (Sec. 6.2). The picture neglects the qS waves, mode conversion, reflections and interference, and it fails where the fibers turn on the wavelength scale. These are exactly the conditions under which we expect, and test for, departures from channeling.

3.3 Predictions

- **W1 (isotropy).** By Proposition 1 and Remark 1, an isotropic lamina makes the operator independent of θ. Every orientation sensitivity vanishes identically. This is an identity of the model, not an empirical prediction; we use it as a consistency check of the discrete gradient.
- **W2 (redirection is the robust effect).** Write the target-energy gain of a design against straight fibers as

$$G = \frac{E_f}{E_f^0} = \underbrace{\frac{E_f / E_{\text{int}}}{E_f^0 / E_{\text{int}}^0}}_{G_\varphi \text{ (redirection)}} \times \underbrace{\frac{E_{\text{int}}}{E_{\text{int}}^0}}_{\text{field growth}}, \tag{9}$$

  where E_f is the energy in the target and E_{int} the energy in the plate interior (outside the sponge). The identity is exact. Channeling acts on G_φ and does not depend on frequency, because (7) is non-dispersive. The field-growth factor depends on standing-wave interference in the finite, lightly damped plate, and so on frequency. A gain beyond G_φ at the design frequency should therefore not survive a frequency change.
- **W3 (flux follows fibers).** In any slowly curving fiber field, optimized or not, the local energy flux should align with the local fiber director far better than with the drive direction, and the ray prediction (8) should reproduce the redirection. The control is a set of curved fields that were never optimized: if channeling were selected by the optimizer rather than imposed by the lamina, the flux would not follow their fibers.
- **W4 (anisotropy).** G_φ should grow with the anisotropy, following s_{20}, from 1 at isotropy.
- **W5 (limits).** A tow of minimum radius R turns by at most ℓ / R over a run ℓ. Energy channeled along the fibers can therefore be redirected only as fast as the fibers turn. Redirection should survive while tows can still turn on the wavelength scale, and fail once R exceeds it. Exact cloaking, by contrast, needs the singular anisotropy of transformation media [19]. A fixed-contrast lamina must therefore rely on interference to cancel a scattered field, and we expect such cancellation to be tuned to one frequency (a hypothesis, tested in Sec. 6.5, not a theorem).

4 Finite-element discretization in `dolfinx`

4.1 Function spaces, element types, and the mesh for each problem

Structured and boundary-conforming meshes both appear, so we state the discrete spaces explicitly before deriving anything. Table 2 collects them.

The state space for the wave-control problems is a *four-component* continuous Lagrange space,

$$\mathsf{U}_h \in \mathcal{W}_h = \left[\mathbb{P}_1(\mathcal{T}_h)\right]^4 = \text{span}\{\texttt{("Lagrange",1,(4,))}\}, \qquad \mathsf{U} = (u_x^r, u_y^r, u_x^i, u_y^i), \tag{10}$$

Table 2: Discrete spaces, element types and meshes ($\mathcal{T}_h$ the cell complex).

	wave control (`dolfinx`)
cells	triangles ($\mathbb{P}_1$ affine)
mesh	structured; `gmsh` conforming w/ holes
state space	$[\mathbb{P}_1]^4$, `("Lagrange",1,(4,))`
DOF per node	4 ($u_x^r, u_y^r, u_x^i, u_y^i$)
DOF per element	12
quadrature	UFL-estimated degree
mass	consistent
design θ	$\mathbb{DG}_0$, `("DG",0)`
parametrization	CS-RBF $\theta = \mathbf{B}\hat{\theta}$
solve	sparse LU (real `PETSc`)
dual	adjoint in $\mathcal{W}_h$, $\mathcal{A}_h^{\mathsf{T}}$ (second sparse LU)

i.e. *four* scalar degrees of freedom per vertex: the two in-plane components of the real part and the two of the imaginary part. This is the discrete carrier of the real four-field split (13) — it is why a real `PETSc` build suffices, and it is the reason the block operator is 4×4 rather than 2×2.

The design fields live in the *discontinuous, element-wise constant* space

$$\theta_h \in \mathcal{S}_h = \mathbb{DG}_0(\mathcal{T}_h) = \operatorname{span}\{\,\texttt{("DG",0)}\,\}, \tag{11}$$

one orientation per cell. The pairing $\mathcal{W}_h = [\mathbb{P}_1]^4$ with $\mathcal{S}_h = \mathbb{DG}_0$ is deliberate: the state must be conforming in H^1, whereas the design must be free to vary cell-by-cell without any inter-element continuity, and the cell gradient (23) is then obtained simply by differentiating the residual against the $\mathbb{DG}_0$ test function.

The adjoint is solved in the same space. We use a Galerkin discretization, so the adjoint state $\mathbf{\Lambda}_h \in \mathcal{W}_h$ — the *same* $[\mathbb{P}_1]^4$ space as the forward state, on the same mesh. Discretely the adjoint operator is the transpose $\mathcal{A}_h^{\mathsf{T}}$ of the forward matrix, not a separately discretized continuous adjoint; this "discretize-then-differentiate" order is what makes the gradient exact for the discrete objective (and is why it matches finite differences to $\sim 10^{-6}$ rather than to truncation error). Practically, the forward solve factors $\mathcal{A}_h = \mathbf{LU}$ and the adjoint solves with the transposed operator $\mathcal{A}_h^{\mathsf{T}}$ by a second sparse LU factorization, so one gradient costs two factorizations (the forward factors could be reused for the transposed solve; we do not exploit this).

Meshes. The lens uses a structured right-triangle mesh of $[0, L_x] \times [0, L_y]$. The cloak and the panel instead use a `gmsh`/OCC mesh in which each circular hole is *cut from the geometry* so the triangulation *conforms* to the circle, refined on the rim. With linear (P1) elements the resolution of the slowest (quasi-shear) wave is about 9 elements per wavelength for the lens ($\omega = 40$), 4.9 for the cloak ($h = 0.045$, $\omega = 50$) and 4.0 for the panel ($h = 0.055$, $\omega = 50$); the coarser two are checked by refinement in Appendix F. The hole boundary is then exactly circular and traction-free by the natural boundary condition of the weak form. We flag the distinction because it is the substantive reason for using `dolfinx` here at all: representing a hole instead as a soft $\mathbb{DG}_0$ density void on a structured grid staircases the boundary to within one element, scatters off the staircase corners, and blurs the very radius R that the curvature rule $\zeta_{\text{all}} = 1/R$ is tied to.

4.2 Real four-field split of the complex vector problem

The weak form of (4) is the complex-symmetric system

$$\big[(1+\mathrm{i}\eta)\mathbf{K}(\boldsymbol{\theta})-\omega^2\mathbf{M}+\mathrm{i}\omega\mathbf{C}\big]\mathbf{u}=\mathbf{f},\quad \mathbf{K}=\int_\Omega \boldsymbol{\varepsilon}(\cdot)^{\mathsf{T}}\mathbf{D}_f(\theta)\boldsymbol{\varepsilon}(\cdot),\ \ \mathbf{M}=\int_\Omega \rho\,(\cdot)\cdot(\cdot),\ \ \mathbf{C}=\int_\Omega c\,(\cdot)\cdot(\cdot). \tag{12}$$

The `PETSc` build stores *real* scalars only, so we split $\mathbf{u}=\mathbf{u}_r+\mathrm{i}\mathbf{u}_i$ (each a 2-vector) and solve the equivalent real block system on the four-component field $U=(u_{xr},u_{yr},u_{xi},u_{yi})$,

$$\begin{bmatrix} A & -B\\ B & A\end{bmatrix}\begin{bmatrix}\mathbf{u}_r\\ \mathbf{u}_i\end{bmatrix}=\begin{bmatrix}\mathbf{f}_r\\ \mathbf{f}_i\end{bmatrix},\qquad A=\mathbf{K}(\boldsymbol{\theta})-\omega^2\mathbf{M},\qquad B=\eta\,\mathbf{K}(\boldsymbol{\theta})+\omega\mathbf{C}, \tag{13}$$

which is exactly $[\mathrm{Re},-\mathrm{Im};\mathrm{Im},\mathrm{Re}]$ of the complex operator (the structural damping $\eta\mathbf{K}$ joins the sponge in the imaginary block B).

Where (13) comes from. Write the complex operator as $\mathcal{A}_{\mathbb{C}}=A+\mathrm{i}B$ with $A=\mathbf{K}-\omega^2\mathbf{M}$ and $B=\eta\mathbf{K}+\omega\mathbf{C}$ *both real and symmetric*. Expanding the complex product and separating real and imaginary parts,

$$(A+\mathrm{i}B)(\mathbf{u}_r+\mathrm{i}\mathbf{u}_i)=\underbrace{\big(A\mathbf{u}_r-B\mathbf{u}_i\big)}_{\text{real part}}+\mathrm{i}\underbrace{\big(B\mathbf{u}_r+A\mathbf{u}_i\big)}_{\text{imaginary part}}\overset{!}{=}\mathbf{f}_r+\mathrm{i}\mathbf{f}_i, \tag{14}$$

and equating the two parts *separately* — legitimate because $\{1,\mathrm{i}\}$ is a basis of $\mathbb{C}$ over $\mathbb{R}$ — gives the two coupled real equations $A\mathbf{u}_r-B\mathbf{u}_i=\mathbf{f}_r$ and $B\mathbf{u}_r+A\mathbf{u}_i=\mathbf{f}_i$, i.e. exactly (13). This is the standard *realification*, the injective ring homomorphism $a+\mathrm{i}b\ \mapsto\ \left[\begin{smallmatrix}a&-b\\ b&a\end{smallmatrix}\right]$ applied blockwise; nothing is approximated, dropped, or symmetrized.

Proposition 2 (Exactness of the split and of its adjoint)**.** *Let $A,B\in\mathbb{R}^{n\times n}$ be symmetric and $\mathcal{A}=\left[\begin{smallmatrix}A&-B\\ B&A\end{smallmatrix}\right]$. Then*

1. *$(\mathbf{u}_r,\mathbf{u}_i)$ solves $\mathcal{A}(\mathbf{u}_r,\mathbf{u}_i)^{\mathsf{T}}=(\mathbf{f}_r,\mathbf{f}_i)^{\mathsf{T}}$ if and only if $\mathbf{u}=\mathbf{u}_r+\mathrm{i}\mathbf{u}_i$ solves $(A+\mathrm{i}B)\mathbf{u}=\mathbf{f}_r+\mathrm{i}\mathbf{f}_i$; in particular $\mathcal{A}$ is invertible exactly when $A+\mathrm{i}B$ is.*
2. *$\mathcal{A}^{\mathsf{T}}=\left[\begin{smallmatrix}A&B\\ -B&A\end{smallmatrix}\right]$ is the realification of $A-\mathrm{i}B=(A+\mathrm{i}B)^{\mathsf{H}}$.*

Proof. (1) is the displayed computation read in both directions. For (2), transposing blockwise and using $A^{\mathsf{T}}=A$, $B^{\mathsf{T}}=B$ gives $\mathcal{A}^{\mathsf{T}}=\left[\begin{smallmatrix}A&B\\ -B&A\end{smallmatrix}\right]$, which is the image of $A-\mathrm{i}B$ under $a+\mathrm{i}b\mapsto\left[\begin{smallmatrix}a&-b\\ b&a\end{smallmatrix}\right]$; and $(A+\mathrm{i}B)^{\mathsf{H}}=A^{\mathsf{H}}-\mathrm{i}B^{\mathsf{H}}=A-\mathrm{i}B$ since A,B are real symmetric. □

Part (2) is the reason the real formulation costs us nothing in the optimization: the discrete adjoint equation (22) is solved with the *transpose* $\mathcal{A}^{\mathsf{T}}$, and Proposition 2 says that transposing the real block system is precisely applying the complex *Hermitian* adjoint. So the real solve returns the true complex adjoint state, not an approximation of it — which is why the resulting gradient matches finite differences to $\sim 10^{-6}$.

Remark 2 (Cost, and when the split can be dropped)**.** *$\mathcal{A}$ is real but* non-*symmetric whenever $B\neq 0$ (i.e. whenever there is damping or a sponge), so it is factored by sparse LU, not Cholesky. The split doubles the unknowns; for a sparse factorization this costs a small constant factor over a native complex build of half the size, which we accept because the split is exact and runs on a real-scalar `PETSc` build. With a complex-scalar `PETSc/SLEPc` the entire construction of this subsection collapses to assembling (12) directly on a two-component complex space, and only the objective (which is already real-valued) needs care. Nothing else in the paper would change: the adjoint, the sensitivity (23) and every reported number are properties of the discrete problem, not of how its scalars are stored.*

We discretize U on the four-component continuous P_1 space (10) and assemble (13) as a single bilinear form. Writing the stiffness pairing $s(\mathbf{a},\mathbf{b}) = \int_\Omega \boldsymbol{\varepsilon}(\mathbf{a})^{\mathsf{T}}\mathbf{D}_f(\theta)\boldsymbol{\varepsilon}(\mathbf{b})$ and the mass/sponge pairings $m(\mathbf{a},\mathbf{b}) = \int_\Omega \rho\,\mathbf{a}\cdot\mathbf{b}$, $d(\mathbf{a},\mathbf{b}) = \int_\Omega c\,\mathbf{a}\cdot\mathbf{b}$ for the test 2-vectors $\mathbf{p}_r, \mathbf{p}_i$,

$$\begin{aligned} a(U,P;\boldsymbol{\theta}) &= s(\mathbf{u}_r,\mathbf{p}_r) - \omega^2 m(\mathbf{u}_r,\mathbf{p}_r) - \eta\, s(\mathbf{u}_i,\mathbf{p}_r) - \omega\, d(\mathbf{u}_i,\mathbf{p}_r) \\ &\quad + s(\mathbf{u}_i,\mathbf{p}_i) - \omega^2 m(\mathbf{u}_i,\mathbf{p}_i) + \eta\, s(\mathbf{u}_r,\mathbf{p}_i) + \omega\, d(\mathbf{u}_r,\mathbf{p}_i), \qquad \ell(P) = \int_\Omega \mathbf{f}\cdot\mathbf{p}_r. \end{aligned} \tag{15}$$

The orientation θ, the sponge c and the region indicators are element-wise (DG_0) fields; $\mathbf{D}_f(\theta)$ is built from (2) directly in UFL. Material, thickness and density are the same everywhere and in every design: voids are cut from the mesh, never represented by a density. We assemble (15) and solve with a sparse LU factorization; write the discrete system $\mathcal{A}(\boldsymbol{\theta})\,\mathsf{U} = \mathsf{b}$.

4.3 Absorbing sponge and drive

The sponge coefficient ramps quadratically into a boundary margin of width m,

$$c(\mathbf{x}) = c_0\left[\max\left(0,\ \tfrac{x-(L_x-m)}{m},\ \tfrac{m-y}{m},\ \tfrac{y-(L_y-m)}{m}\right)\right]^2, \tag{16}$$

absorbing outgoing waves at the right, top and bottom edges (the left edge carries the drive). The layer is imperfect: for the along-fiber qP wave it reflects $|R| = 0.31$–0.34 at the lens frequencies, and $|R| \le 0.13$ when its width is doubled (Appendix F), and the finite plate is resonant (the resonant growth persists with the better absorber and scales as $1/\eta$, Appendix F, so it is not set by the absorber alone). The drive is an x-directed line source concentrated near the left edge,

$$\mathbf{f}(\mathbf{x}) = \left(A\,e^{-\beta(x-x_0)^2},\ 0\right), \qquad (A,\beta,x_0) = (30,\ 80,\ 0.1), \tag{17}$$

a plane pressure wave. Parameters (c_0, m) are $(80, 0.6)$ for the lens and $(70, 0.6)$ for the cloak (Table 8). Since $\mathbf{C}, \mathbf{M}$ and $\mathbf{f}$ are θ-independent, they drop out of every design sensitivity (§5).

5 Objectives and the discrete-adjoint sensitivity

Both objectives are quadratic in the state, $J = \int_\Omega w\,\Psi(U)$ with w a region indicator:

$$\Psi_{\text{lens}} = |\mathbf{u}|^2 = \sum_{k=0}^{3} U_k^2, \qquad \Psi_{\text{cloak}} = |\mathbf{u}-\mathbf{u}_{\text{ref}}|^2 = \sum_{k=0}^{3} (U_k - U_k^{\text{ref}})^2, \tag{18}$$

the focus energy in a target box (maximized) and the far-field mismatch to a void-free reference (minimized). Throughout, "energy" in a region means this displacement-squared measure $\int |\mathbf{u}|^2$, which for a time-harmonic field is proportional to the time-averaged kinetic energy.

5.1 Energy flux, power balance and the power routed into a region

For the convention $\mathrm{Re}\{\mathbf{u}e^{+\mathrm{i}\omega t}\}$ the velocity is $v = \mathrm{i}\omega\mathbf{u}$, and the time-averaged energy flux is $P_j = -\frac{1}{2}\,\mathrm{Re}\{\sigma_{ij}\bar{v}_i\}$. With $\mathbf{u} = \mathbf{u}^{\mathrm{r}} + \mathrm{i}\mathbf{u}^{\mathrm{i}}$ and the elastic stress $\boldsymbol{\sigma} = \mathbf{D}_f\boldsymbol{\varepsilon}$,

$$P_j = \tfrac{\omega}{2}\left[\sigma_{ij}(\mathbf{u}^{\mathrm{r}})\,u_i^{\mathrm{i}} - \sigma_{ij}(\mathbf{u}^{\mathrm{i}})\,u_i^{\mathrm{r}}\right], \tag{19}$$

evaluated per cell from the solved field. We use the elastic part of the stress; the hysteretic part $i\eta \mathbf{D}_f \boldsymbol{\varepsilon}$ changes $\boldsymbol{P}$ by a relative $O(\eta) = 2\%$. The time-averaged powers of the discrete system satisfy an exact balance,

$$P_{\text{in}} = -\tfrac{\omega}{2}\int \mathbf{f}\cdot\mathbf{u}^{\text{i}} = \underbrace{\tfrac{\omega\eta}{2}\int \left[\boldsymbol{\varepsilon}(\mathbf{u}^{\text{r}})^{\mathsf{T}}\mathbf{D}_f\boldsymbol{\varepsilon}(\mathbf{u}^{\text{r}}) + \boldsymbol{\varepsilon}(\mathbf{u}^{\text{i}})^{\mathsf{T}}\mathbf{D}_f\boldsymbol{\varepsilon}(\mathbf{u}^{\text{i}})\right]}_{P_\eta \text{ (structural loss)}} + \underbrace{\tfrac{\omega^2}{2}\int c\,|\mathbf{u}|^2}_{P_c \text{ (sponge)}}, \tag{20}$$

which our solutions satisfy to 10^{-13}. In steady state the net power flowing into a region without sponge equals the power dissipated inside it. The share of the injected power delivered to the target is therefore $\Pi = P_{\eta,\text{target}}/P_{\text{in}}$, and its ratio to straight fibers, Π/Π_0, measures routing of *power*, as opposed to the redistribution of stored energy measured by G_φ (Sec. 3.3).

5.2 Lagrangian, adjoint equation, and residual derivative

Adjoin the forward problem $\mathcal{A}(\boldsymbol{\theta})\mathsf{U} - \mathsf{b} = \mathbf{0}$ with a multiplier $\boldsymbol{\Lambda}$: $\mathcal{L} = J(\mathsf{U}) + \boldsymbol{\Lambda}^{\mathsf{T}}(\mathcal{A}\mathsf{U} - \mathsf{b})$. On the constraint manifold $\mathcal{L} = J$, and expanding the total derivative,

$$\frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \underbrace{\left(\tfrac{\partial J}{\partial \mathsf{U}} + \mathcal{A}^{\mathsf{T}}\boldsymbol{\Lambda}\right)^{\mathsf{T}}\tfrac{\mathrm{d}\mathsf{U}}{\mathrm{d}\theta_\ell}}_{\text{eliminated by the adjoint}} + \boldsymbol{\Lambda}^{\mathsf{T}}\frac{\partial \mathcal{A}}{\partial \theta_\ell}\mathsf{U}. \tag{21}$$

Choosing $\boldsymbol{\Lambda}$ to kill the implicit state term gives the *discrete adjoint equation*, and the design derivative then reduces to the explicit θ-dependence of the residual:

$$\boxed{\mathcal{A}(\boldsymbol{\theta})^{\mathsf{T}}\boldsymbol{\Lambda} = -\frac{\partial J}{\partial \mathsf{U}}} \qquad \boxed{\frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \frac{\partial R}{\partial \theta_\ell}, \quad R(\boldsymbol{\theta}) \equiv a(\mathsf{U},\boldsymbol{\Lambda};\boldsymbol{\theta}) - \ell(\boldsymbol{\Lambda})} \tag{22}$$

with $\mathsf{U}, \boldsymbol{\Lambda}$ frozen. Only the stiffness terms in R carry θ, so

$$\frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \int_{\Omega_\ell}\Big[\boldsymbol{\varepsilon}(\mathbf{u}_r)^{\mathsf{T}}\tfrac{\partial \mathbf{D}_f}{\partial\theta}\boldsymbol{\varepsilon}(\boldsymbol{\lambda}_r) + \boldsymbol{\varepsilon}(\mathbf{u}_i)^{\mathsf{T}}\tfrac{\partial \mathbf{D}_f}{\partial\theta}\boldsymbol{\varepsilon}(\boldsymbol{\lambda}_i) + \eta\big(\boldsymbol{\varepsilon}(\mathbf{u}_r)^{\mathsf{T}}\tfrac{\partial \mathbf{D}_f}{\partial\theta}\boldsymbol{\varepsilon}(\boldsymbol{\lambda}_i) - \boldsymbol{\varepsilon}(\mathbf{u}_i)^{\mathsf{T}}\tfrac{\partial \mathbf{D}_f}{\partial\theta}\boldsymbol{\varepsilon}(\boldsymbol{\lambda}_r)\big)\Big], \tag{23}$$

where $\boldsymbol{\lambda}_r, \boldsymbol{\lambda}_i$ are the real/imag adjoint displacements and $\partial\mathbf{D}_f/\partial\theta = \mathbf{T}'\mathbf{Q}_f\mathbf{T}^{\mathsf{T}} + \mathbf{T}\mathbf{Q}_f\mathbf{T}'^{\mathsf{T}}$. The mass, sponge and source carry no θ-dependence and drop out: the whole gradient is built from $\partial\mathbf{D}_f/\partial\theta$, so it vanishes identically for an isotropic material, making Remark 1 an exact statement about the gradient.

5.3 Automatic implementation, scaling, and verification

Each boxed step is one line of UFL. The adjoint load $\partial J/\partial\mathsf{U}$ is `derivative`$(w\Psi(U)\,\mathrm{d}\mathbf{x}, U, P)$; (22) is solved with the transposed operator $\mathcal{A}^{\mathsf{T}}$ (a second sparse LU factorization); the cell gradient is `derivative`$(R,\theta,\hat{P})$ against the DG_0 test function — no element derivatives by hand. Both the lens and cloak gradients match central finite differences to a relative error $\sim 10^{-6}$. The raw objectives are tiny ($\sim 10^{-4}$), so for the lens we descend the scale-invariant *log*-objective $g_\ell = (\partial_{\theta_\ell} J)/J$, restoring $O(1)$ gradients for the Method of Moving Asymptotes (MMA) [40] under $\theta_\ell \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

5.4 Smooth orientation parametrization and manufacturable toolpaths

A raw per-cell orientation is mesh-scale noisy — un-manufacturable and, driving the wave field, incoherent. We therefore parametrize θ by a coarse grid of support-point angles $\hat{\theta} = \{\hat{\theta}_i\}$ at sites $\{\mathbf{s}_i\}$ (spacing h_s) interpolated to the cell centroids $\{\mathbf{x}_c\}$ with a compactly supported Wendland C^2 radial basis [41] $\phi(q) = (1-q)^4(4q+1)$ (support $q < 1$), as in [11]:

$$\theta_c = (B\hat{\theta})_c = \sum_i B_{ci}\hat{\theta}_i, \qquad B_{ci} = \frac{\phi(q_{ci})}{\sum_j \phi(q_{cj})}, \qquad q_{ci} = \frac{\|\mathbf{x}_c - \mathbf{s}_i\|}{r}, \tag{24}$$

a row-stochastic map ($\sum_i B_{ci} = 1$); B^{T} propagates to the gradient, $\partial J/\partial\hat{\theta} = B^{\mathsf{T}}(\partial J/\partial\theta)$. Support spacing and radius are $(h_s, r) = (0.33, 0.7)$ for the lens and $(0.28, 0.6)$ for the cloak. This CS-RBF map makes the design low-dimensional and smooth *by construction*. When the problem has a mirror symmetry (as the lens does about $y = L_y/2$) we enforce it on the design: a fiber field is symmetric under $y \mapsto L_y - y$ iff $\theta(x, L_y - y) = -\theta(x, y)$, so we use a support grid symmetric about the axis and project the design each iteration onto that subspace, $\hat{\theta} \mapsto \frac{1}{2}(\hat{\theta} - \hat{\theta}_{\text{mirror}})$. This removes the spurious asymmetry that the (non-symmetric) triangulation and optimizer would otherwise introduce, yielding a symmetric lens with a horizontal fiber on the axis. We delineate the resulting fiber *toolpaths* with the wave projection of [11]: from the sign-free double-angle director $(\cos 2\theta, \sin 2\theta)$ we recover a smooth (seam-free) orientation branch by region-growing sign propagation, then solve for a phase field Φ whose gradient is unit and orthogonal to the fibers,

$$\nabla\Phi(\mathbf{x}) = \mathbf{g} \equiv [-\sin\theta,\ \cos\theta]^{\mathsf{T}}, \qquad \Phi = \arg\min_{\Phi} \int_\Omega \|\nabla\Phi - \mathbf{g}\|^2, \tag{25}$$

discretized as a sparse least-squares gradient match on the grid (`lsqr`; matching $\nabla\Phi$ *directly* keeps $|\nabla\Phi|$ close to 1 wherever the director field is nearly divergence-free — the normal-equations Poisson form drops the boundary flux and collapses the pitch). The level sets of Φ at increments d — equivalently the peaks of the wave projection $\chi = \frac{1}{2} + \frac{1}{2}\cos(2\pi\Phi/d)$ — are the as-manufactured tow center-lines; they follow the fiber direction, are masked out of any prescribed void, so no toolpath crosses a hole, and their spacing is d only where the field does not converge or diverge: where it converges (as near the axis of the lens, Fig. 2c) the tows crowd together, which a fiber placement process would resolve by cutting or adding tows. These paths visualize the field; we do not model the resulting thickness or gap variations. Together with the structural damping η, this smoothing is what renders the optimized fields coherent rather than speckled.

5.5 Fiber-curvature (curl) constraint

A printable toolpath cannot bend arbitrarily tightly: the tow has a minimum turning radius. Following [11], we bound the local *fiber-path curvature* — the directional derivative of the orientation along the fiber, $\mathbf{n} = (\cos\theta, \sin\theta)$,

$$\zeta(\mathbf{x}) = (\mathbf{n}\cdot\nabla)\theta = \cos\theta\,\partial_x\theta + \sin\theta\,\partial_y\theta, \qquad |\zeta(\mathbf{x})| \le \zeta_{\text{all}}, \tag{26}$$

so ζ is exactly the signed curvature of the fiber streamlines and ζ_{all} is the inverse minimum radius. The spatial derivatives come *analytically* from the CS-RBF map (24): $\partial_x\theta = \mathbf{B}_x\hat{\theta}$, $\partial_y\theta = \mathbf{B}_y\hat{\theta}$. Writing $W_{ci} = \phi(q_{ci})$, $S_c = \sum_j W_{cj}$ (so $B_{ci} = W_{ci}/S_c$) and differentiating the kernel *and* its normalization consistently,

$$\partial_x W_{ci} = -\frac{20(1-q_{ci})^3}{r^2}(x_c - s_{ix}), \qquad (\mathbf{B}_x)_{ci} = \frac{\partial_x W_{ci}}{S_c} - B_{ci}\frac{\sum_j \partial_x W_{cj}}{S_c}, \tag{27}$$

and likewise $\mathbf{B}_y$ (using $\phi'(q) = -20\,q(1-q)^3$); we impose (26) through a penalty on the cell average of the excess of ζ^2 over ζ^2_{all},

$$\mathcal{C}(\hat{\theta}) = \left\langle \max\!\left(0,\ \zeta^2 - \zeta^2_{\text{all}}\right) \right\rangle, \qquad \min_{\hat{\theta}} \ -\ln E(\hat{\theta}) + \mu\,\mathcal{C}(\hat{\theta}), \tag{28}$$

whose gradient uses the curvature Jacobian $\partial\zeta/\partial\hat{\theta} = \operatorname{diag}(-\sin\theta\,\theta_x + \cos\theta\,\theta_y)\mathbf{B} + \operatorname{diag}(\cos\theta)\mathbf{B}_x + \operatorname{diag}(\sin\theta)\mathbf{B}_y$ (finite-difference verified to $\sim 10^{-10}$), with the penalty weight μ raised by continuation until the constraint is met.

6 Results

All results use the CFRP lamina of Sec. 2 unless stated, an x-directed drive at the left edge (a plane-wave line source for the lens and the cloak, a Gaussian beam for the panel, Table 9), and the absorbing sponge on the other three sides. Only the fiber orientation is designed. Material, thickness and mass are identical to the straight-fiber baseline.

6.1 The lens: what the optimizer buys, and what it does not (W1, W2)

On a 4×3 plate at $\omega = 40$ (Table 8), maximizing the energy in a target box (dashed square in Fig. 2) yields a mirror-symmetric field of tows bending toward the target near the right end of the plate. Its single-frequency target-energy gain against straight fibers is $G = 78\times$. The decomposition (9) shows what this number contains (Fig. 3, Table 3).

The benchmark: a converged absorbing layer. A finite plate with a reflecting boundary holds standing waves, and reflections from the layer feed the target. We therefore lead with an absorbing layer whose reflection is converged. The paper's layer (width 0.6) reflects $|R| \approx 0.31$–0.34 of the along-fiber qP wave. Widening it outward, with the interior unchanged, gives $|R| \le 0.13$, 0.04 and 0.015 for widths 1.2, 1.8 and 2.4 (`wt_r1_lens.py absorber`). Over that series the band-averaged redirection, both of the paper's lens and of a lens re-optimized with the wider layer, is 3.9–4.0$\times$ and no longer changes with $|R|$ (Fig. 3b). This is the absorber-independent redirection of the lens. With the paper's layer the same lens gives 5.2$\times$: reflections inflate the redirection by about a third. The band average $\langle G_\varphi \rangle_{\text{band}}$ is defined throughout as the mean of G_φ at $\omega \in \{35, 40, 45, 50\}$.

Redirection, power, and growth.

- **Redirection is broadband over the tested wavelengths.** The target's share of the interior energy rises from 1.47% to 8.6% at $\omega = 40$ ($G_\varphi = 5.8\times$; with the wider layer, re-optimized, 4.4$\times$). Sweeping the drive frequency on the *fixed* design, $G_\varphi \ge 3.7\times$ for every $\omega \in [37.5, 60]$ ($\ge 3.6\times$ over $[40, 60]$ with the wider layer). At $\omega = 20$ and 25 the redirection collapses and even reverses ($G_\varphi < 1$): there the along-fiber wavelength ($\lambda_\parallel \ge 2.3$) exceeds the length over which the fibers turn, and the ray picture no longer applies. "Broadband" therefore means $\lambda_\parallel \lesssim 1.5$ for this lens, not every frequency.
- **Power is routed, not only stored.** The share of the injected power that the target absorbs (Sec. 5.1) rises from 0.35% to 2.7% at $\omega = 40$, and its ratio to straight fibers is 3.8–7.7$\times$ over $\omega \in [35, 50]$ (3.2–12.8$\times$ for the re-optimized lens with the wider layer; Fig. 3d). Stored-energy share and absorbed-power share agree in sign and order of magnitude at every frequency tested, so the redirection is a redirection of power flow, not only of standing energy.

Table 3: The lens design (optimized at $\omega = 40$, paper's absorbing layer) evaluated across frequency. G: target-energy gain; G_φ: redirection (target share of interior energy, relative to straight fibers); growth: interior energy ratio; P: injected power ratio. $G = G_\varphi \times$growth exactly.

ω	G	G_φ	growth	P/P_0
20.0	0.2	0.25	0.74	0.61
25.0	0.9	0.41	2.26	0.96
30.0	3.7	1.65	2.25	0.92
35.0	2.1	2.34	0.89	0.49
37.5	6.6	3.74	1.77	0.80
40.0	**78.0**	**5.82**	**13.41**	**4.13**
42.5	10.1	6.10	1.65	0.95
45.0	16.3	6.54	2.49	1.28
50.0	5.5	6.01	0.91	0.70
55.0	5.7	4.27	1.34	0.91
60.0	5.8	3.71	1.58	1.08

- **The excess gain is whole-field growth, consistent with resonance.** The interior field grows 13.4× at $\omega = 40$ (15.9× with the wider layer, where $G = 70\times = 4.4\times$ redirection $\times$ 15.9× growth). A fine sweep ($\Delta\omega = 0.25$) shows a single peak of the growth, 13.6× at $\omega = 39.8$, with a half-power width of 1.6, i.e. a quality factor ≈ 26, consistent with the plate's loss factor $\eta = 0.02$ ($Q = 50$) plus the sponge loss. Across the peak G_φ varies smoothly (3.0–7.0) without a peak. The growth also scales as $1/\eta$ while G_φ does not (Appendix F). At $\omega = 37.5$ and 42.5, only ±6% away, G falls to 6.6× and 10.1×.

A large single-frequency focus gain is therefore not by itself evidence of steering. It mostly measures how well the optimizer tuned a standing wave. The transferable result is the redirection.

For W1, setting the lamina isotropic makes $G = G_\varphi = 1$ to machine precision, and every orientation gradient vanishes identically (Remark 1; Sec. 6.3).

6.2 Mechanism: the energy flux follows the fibers (W3)

We compute the time-averaged energy flux (19) cell by cell and its angle to the local fiber director (Fig. 4). Cells whose flux is below 5% of the maximum are excluded: there the direction is ill-defined (standing-wave nodes), and they carry little power. The remaining 66% of the interior carries the energy transport.

- **Optimized lens.** The $|\boldsymbol{P}|$-weighted mean deviation of the flux from the fibers is 5.8° at $\omega = 40$ (6.7° without the mask), and 94% of the flux runs within 20° of them. The flux deviates 19.2° on average from the drive direction x. Across the band the deviation is 11.6°, 5.5° and 4.5° at $\omega = 35, 45, 50$.
- **Where channeling fails.** The deviation map (Fig. 4b) is small almost everywhere and large in one place: just behind the target, where the fibers converge sharply. Binned by how fast the fibers turn (Fig. 4d), the mean deviation stays at 2–4° while the fibers turn by less than about 4 rad per along-fiber wavelength, and rises to 12–24° beyond. This is the ray-validity condition of Sec. 3 made quantitative. The lens's sharpest turning has a gradient length $L_\theta = 0.15$, a tenth of $\lambda_\parallel$ (Table 1).
- **qP content.** 71% of the lens's interior kinetic energy is polarized along the local fibers (100% with straight fibers; 52–79% across the band). In CFRP the qP polarization lies within a few

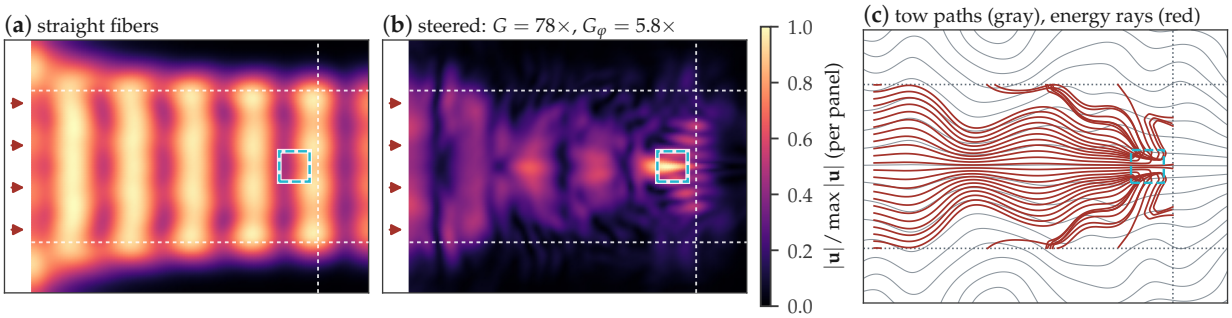


Figure 2: **The orientation-only lens concentrates the field on its target, and rays traced through its fibers converge there.** Full 4×3 plate at the design frequency $\omega = 40$, driven along x at its left edge (arrows); dotted white lines: where the absorbing layer begins; dashed square: the target $(0.78L_x, L_y/2) \pm 0.18$; structured mesh, 9 elements per shortest (qS) wavelength. **(a)** Field magnitude $|\mathbf{u}|$ with straight fibers and **(b)** with the optimized fiber field, each normalized by its own maximum, so the comparison is of *where* the energy goes; the gain and redirection are in the title. **(c)** Optimized tow paths (gray) and qP energy rays traced through them by (8) from the drive (red), absorbed where the layer begins.

degrees of the fiber for most wave normals, so this is the qP-like share. The rest is qS motion and mode conversion, which channeling does not describe.

- **Rays predict the redirection.** Tracing qP rays (8) through the fiber field alone (`wt_r1_rays.py`; Fig. 2c) gives a target share of 10.3% against 2.2% for straight fibers. That is $G_{\text{ray}} = 4.7\times$ for the paper's lens and $4.2\times$ for the re-optimized one, against 3.9–$4.0\times$ from the full field with a converged absorber. The ray picture, which has no reflections, interference or resonance, accounts for the measured redirection to within about 20%.
- **Control: fields that were never optimized.** Three random smooth CS-RBF fields (same support grid as the lens, support values drawn from $\mathcal{N}(0, 0.5\,\text{rad})$) and one analytic field, $\theta = 0.6\sin(\pi x/2)\sin(2\pi y/3)$, give flux–fiber deviations of only 1.0–4.2° at $\omega = 40$ and 50, against 9–15° from the drive direction (Appendix F). Their fields turn more gently than the lens's, which is why they channel more tightly.

The energy goes where the fibers point, at every frequency tested and whether or not the field was optimized, wherever the fibers turn slowly on the wavelength scale. Channeling is imposed by the lamina; the optimizer only chooses where the fibers point. This is why the redirection is broadband.

6.3 How the control scales with anisotropy (W4)

Along the family $\mathbf{Q}(s) = \mathbf{Q}_{\text{iso}} + s(\mathbf{Q}_f - \mathbf{Q}_{\text{iso}})$, which holds the isotropic invariants and scales the anisotropic ones by s, we re-optimize the lens for each material from straight fibers and from three random starts (`wt_r1_dose_ms.py`), and evaluate every design with both the paper's layer and the wider one (Table 4, Fig. 5a). The spread over starts is up to 1.3 in $\langle G_\varphi \rangle$. At $E_{f1}/E_{f2} = 2.9$, for example, the four starts give 2.9–4.2, and the straight-start run is the highest of them. The best of four rises monotonically with anisotropy under both absorbing layers: from exactly 1 at isotropy

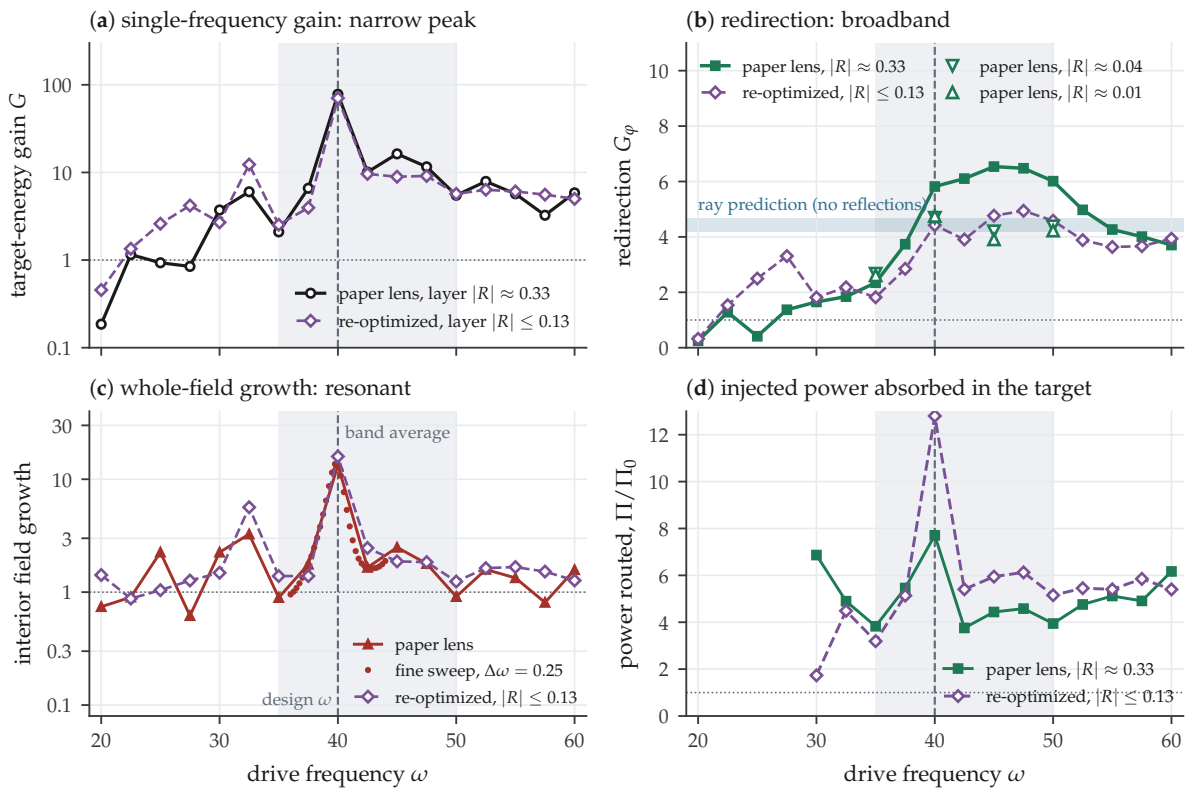


Figure 3: **The lens's single-frequency gain is a narrow resonance; its redirection and its routing of power are broadband and survive a converged absorbing layer.** All on one frequency axis; dashed vertical line: the design frequency $\omega = 40$; shaded: the band over which $\langle G_\varphi \rangle$ is averaged; dotted: parity with straight fibers. Solid green: the paper's lens with the paper's layer ($|R| \approx 0.31$–0.34); dashed purple: the lens re-optimized with a layer twice as wide ($|R| \leq 0.13$). **(a)** Target-energy gain G. **(b)** Redirection G_φ; open triangles: the paper's lens with layers of width 1.8 and 2.4 ($|R| \approx 0.04$ and 0.015); blue band: the ray prediction from the two fiber fields (Sec. 6.2). **(c)** Interior field growth, with a fine sweep ($\Delta\omega = 0.25$) through the peak. **(d)** Share of the injected power absorbed in the target, relative to straight fibers (20).

to 2.7× at $E_{f1}/E_{f2} = 1.7$, 4.2× for CFRP and 5.0× at $E_{f1}/E_{f2} = 32$ with the wider layer (5.7× for CFRP with the paper's). The single-frequency gain G along the same family is *non-monotone* (24× at $E_{f1}/E_{f2} = 2.9$ but 16× at 5.4), because it is dominated by whichever resonance each optimization tunes. W4 holds for the best design found. The detailed shape of the curve is within the spread of the optimizer.

6.4 Manufacturing limits (W5): tow curvature

We bound the tow curvature $|\zeta| \leq \zeta_{\text{all}} = 1/R$, keeping the lens's own mesh, frequency and parametrization (Table 5, Fig. 5b), and state R in along-fiber wavelengths $\lambda_{\parallel} = 1.42$ (Table 1).

- **Plateau.** The best band-averaged redirection stays at 6.4–6.8× for every limit down to $\zeta_{\text{all}} = 1.0$, i.e. a minimum tow radius $R = 1.0 \approx 0.7\lambda_{\parallel}$ (against the cross-fiber qP wavelength 0.37 the same radius is 2.7 wavelengths). We use the along-fiber value because the

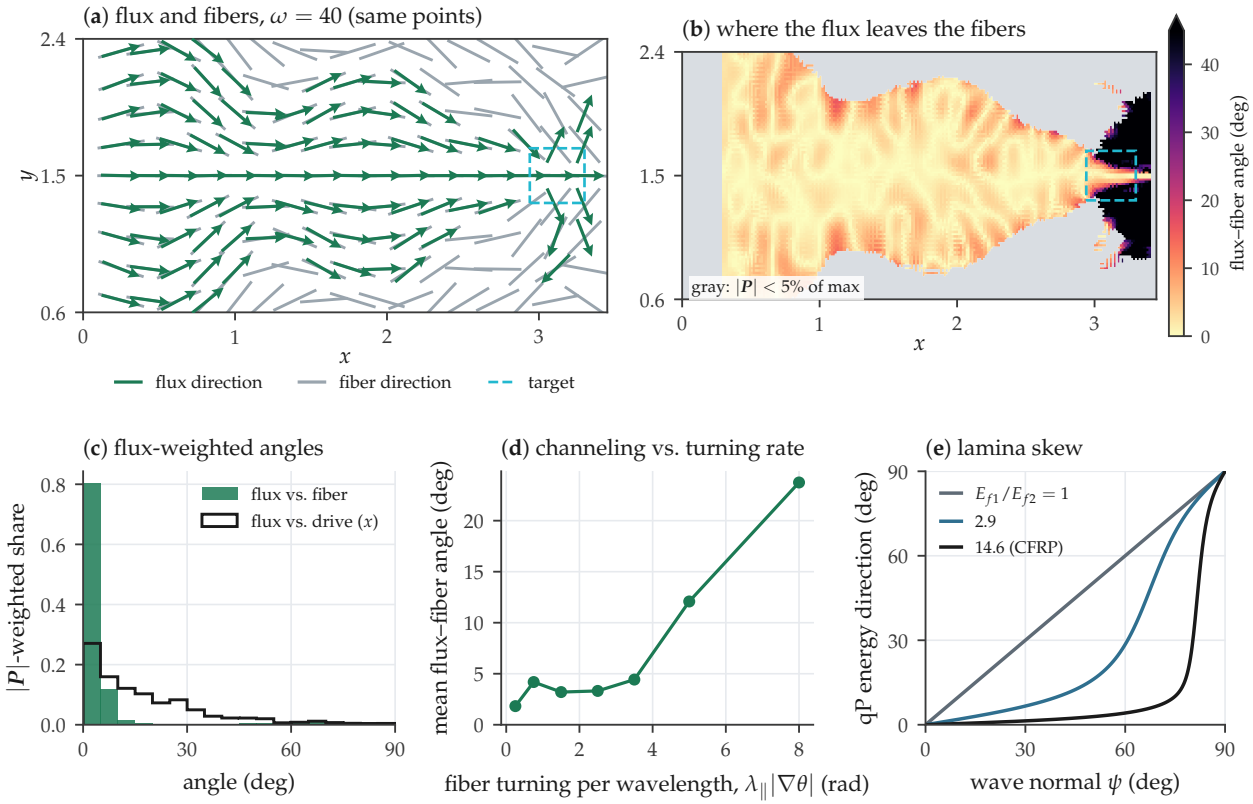


Figure 4: **The energy flux follows the fibers wherever they turn slowly on the wavelength scale.** Optimized lens at $\omega = 40$, window $0 \le x \le 3.4$, $0.6 \le y \le 2.4$ (excluding the absorbing layers). **(a)** Flux direction (green arrows) and fiber direction (gray segments) at the same points; arrows are omitted where $|\boldsymbol{P}| < 5\%$ of its maximum. **(b)** Angle between flux and fiber; gray: $|\boldsymbol{P}| < 5\%$ of its maximum (direction ill-defined). **(c)** $|\boldsymbol{P}|$-weighted distribution of the flux–fiber angle and of the flux–drive angle (strong-flux cells). **(d)** Mean flux–fiber angle against the local fiber turning per along-fiber wavelength, $\lambda_{\parallel}|\nabla\theta|$. **(e)** qP energy direction vs. wave normal (7) for an isotropic lamina, the intermediate lamina $E_{f1}/E_{f2} = 2.9$ of the family $\mathbf{Q}(s)$, and CFRP.

Table 4: Lens re-optimized for each lamina of the anisotropy family. s_{20}: channeling share (Sec. 3); $G(40)$, $G_\varphi(40)$: the straight-start run; last two columns: range of $\langle G_\varphi \rangle_{\text{band}}$ over four starts with the paper's absorbing layer and with the layer twice as wide.

s	E_{f1}/E_{f2}	s_{20}	$G(40)$	$G_\varphi(40)$	$\langle G_\varphi \rangle_{\text{band}}$, 4 starts	better absorber, 4 starts
0.00	1.0	22%	1.0	1.00	1.00	1.00
0.25	1.7	42%	6.9	2.63	1.86–1.88	2.68–2.71
0.50	2.9	60%	24.1	6.92	2.94–4.21	2.09–3.00
0.75	5.4	75%	16.2	5.49	3.59–4.62	2.61–3.35
1.00	14.6	88%	78.0	5.82	5.13–5.74	3.92–4.25
1.10	31.9	94%	162.4	5.85	6.06–6.94	4.46–4.95

Table 5: Lens tow-curvature front (same mesh, frequency and parametrization as Sec. 6.1). Each row is the best band-averaged redirection over all computed designs whose curvature satisfies the limit (a lower bound on what the limit allows); the last column is the range over the independent starts run at that limit.

ζ_{all}	$R = 1/\zeta_{\text{all}}$	$\max\lvert\zeta\rvert$	$G(40)$	$G_\varphi(40)$	$\langle G_\varphi\rangle_{\text{band}}$	range over starts
∞	0	1.47	13.2	7.49	6.80	–
4	0.25	1.47	13.2	7.49	6.80	–
2.5	0.40	1.47	13.2	7.49	6.80	–
1.5	0.67	1.47	13.2	7.49	6.80	1.74–6.80
1	1.00	0.98	11.1	7.08	6.43	1.04–6.43
0.8	1.25	0.76	7.3	5.28	4.87	0.61–4.87
0.6	1.67	0.57	4.5	3.42	3.21	0.69–3.21
0.45	2.22	0.43	3.1	2.48	2.31	0.76–2.31
0.3	3.33	0.28	2.2	1.89	1.75	0.84–1.75

channeled energy travels along the fibers. The front lies at the edge of the slowly turning regime assumed by the ray argument, so we read it as an empirical scale rather than a ray prediction.

- **Decline.** Beyond that radius the best design found falls gradually: 4.9× at $R = 1.25$, 3.2× at $R = 1.67$ ($\approx 1.2\lambda_{\parallel}$), 2.3× at $R = 2.2$ and 1.75× at $R = 3.33$. Tows that cannot turn on the wavelength scale cannot converge the channeled energy on the target within the plate.

How the front is computed, and how robust it is. The optimizations are local and strongly start-dependent under a curvature limit. At each limit from $\zeta_{\text{all}} = 0.3$ to 1.5 we therefore run 8 starts (7 of which lead to distinct designs): every earlier front design and three random fields, each scaled until feasible (`lens_front_robust.py`). The spread is large (at $\zeta_{\text{all}} = 1.0$ from 1.0× to 6.4×; shaded bars in Fig. 5b), so single runs can stall far below what the limit permits. A design whose actual curvature satisfies a limit is feasible for that limit and for every looser one, so Table 5 reports, at each limit, the best band-averaged redirection over *all* computed designs that satisfy it; for $\zeta_{\text{all}} = \infty$ every design counts. These values are lower bounds on what each limit allows. They are with the paper's absorbing layer; by Sec. 6.1 the absorber-independent values are about a quarter lower, and the shape of the front is not affected.

6.5 The cloak: what bounded anisotropy can and cannot do (W5)

The cloak is a harder and less robust problem than the lens, and we report it with that caveat. On an 8×5 plate at $\omega = 50$, a circular traction-free void of radius $R_V = 0.55$ is cut conformally from the mesh. Orientation may change only inside a shell $r < R_{\text{des}} = 1.5$ (exactly straight outside, enforced by a smooth taper carried through the chain rule). The far-field mismatch to the void-free (incident) field is minimized outside $R_C = 1.7$, a region the design cannot touch.

- **At the design frequency, on the design mesh.** The optimized shell reduces the mismatch 36×. Fig. 6e–h shows what this means: the *scattered* field $|\mathbf{u} - \mathbf{u}_{\text{inc}}|$, on the same scale as the incident field, is nearly extinguished outside R_C while it remains strong inside the shell, so the cancellation is a far-field effect, not a quiet plate. The transformation-cloak prescription [17, 19], imposed as fiber *directions* on our fixed-contrast lamina, gives 1.9×, because what

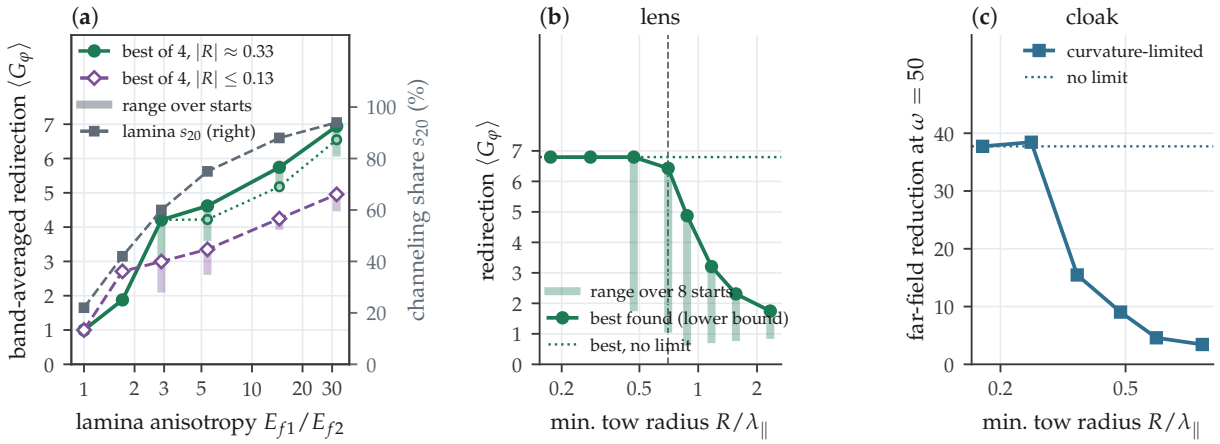


Figure 5: **Redirection grows with anisotropy and survives until the minimum tow radius exceeds about** 0.7 **along-fiber wavelengths. (a)** Band-averaged redirection of the re-optimized lens vs. lamina anisotropy: best of four starts with the paper's layer (green) and with the layer twice as wide (purple), with the range over the starts (bars), the straight-start run (open circles, dotted), and the channeling share s_{20} of the lamina (right axis). **(b)** Band-averaged redirection of the lens and **(c)** far-field reduction of the cloak at $\omega = 50$ vs. the minimum tow radius in along-fiber qP wavelengths ($\lambda_{\parallel} = 1.42$ and 1.14). In (b), circles: the best design found at each limit over all computed designs that satisfy it (a lower bound); shaded bars: the range over the independent starts; dotted: the best design found with no limit; dashed: $R = 0.7\lambda_{\parallel}$. In (c), dotted: the loosest-limit run of this study ($\zeta_{\text{all}} = 5$, effectively unconstrained). At $\zeta_{\text{all}} = 3.5$ the constrained cloak (38.4×) slightly exceeds the unconstrained one (37.7×), within optimizer variability.

it actually requires is a diverging *magnitude* of anisotropy, which no lamina has. Its fiber curvature is also singular at the rim.

- **Mesh sensitivity of a fixed design.** An interference cancellation is sensitive to the discrete dispersion of the mesh it was tuned on. Transferred unchanged to conforming meshes with $h = 0.045$, 0.0375, 0.03 and 0.025 (53,722 to 173,807 cells), the same design gives 36, 33, 26 and 24×. The series has not converged. The curvature-limited design changes less (9.6 → 8.0×) and the transformation directions hardly at all (1.9 → 2.1×). Re-optimized on the $h = 0.03$ mesh (warm start, 80 iterations), the shell reaches 40× and is just as narrowband (7.4× and 14.6× at ±5%). The capability persists under refinement, while any one optimized design is tuned to its discretization. Its converged value is therefore not established, and we give the cloak correspondingly less weight than the lens.
- **Absorbing layer.** With a layer twice as wide (domain enlarged outward, interior unchanged) the three designs give 34, 9.3 and 2.5× (`wt_r1_cloak.py sponge`), so the cancellation is not an artifact of reflections from the layer.
- **Across frequency.** As W5 predicts, the optimized cloak is narrowband (Table 6): 7.5× at $\omega = 47.5$, 14.9× at 52.5, and 1.9× at 40. There it is outperformed by the transformation-direction field (4.6×).
- **Under a tow-curvature limit.** With the curvature bounded at the void's own scale, $\zeta_{\text{all}} = 1/R_V$, the cloak reaches 9.6× at the design frequency. This is less peaked but also less narrowband (5.7–6.2× over ±5%, and at least 2.9× over $\omega \in [40, 60]$), and less mesh-sensitive.

Table 6: Far-field mismatch reduction of three cloak fields across frequency (all on the paper's conforming mesh, $h = 0.045$; designs fixed).

ω	optimized	curvature-limited ($1/R_V$)	transformation directions
40.0	1.9	3.7	4.6
45.0	2.5	2.9	1.8
47.5	7.5	6.2	2.8
50.0	**36.2**	**9.6**	**1.9**
52.5	14.9	5.7	3.9
55.0	6.4	4.6	2.6
60.0	2.7	2.9	1.1

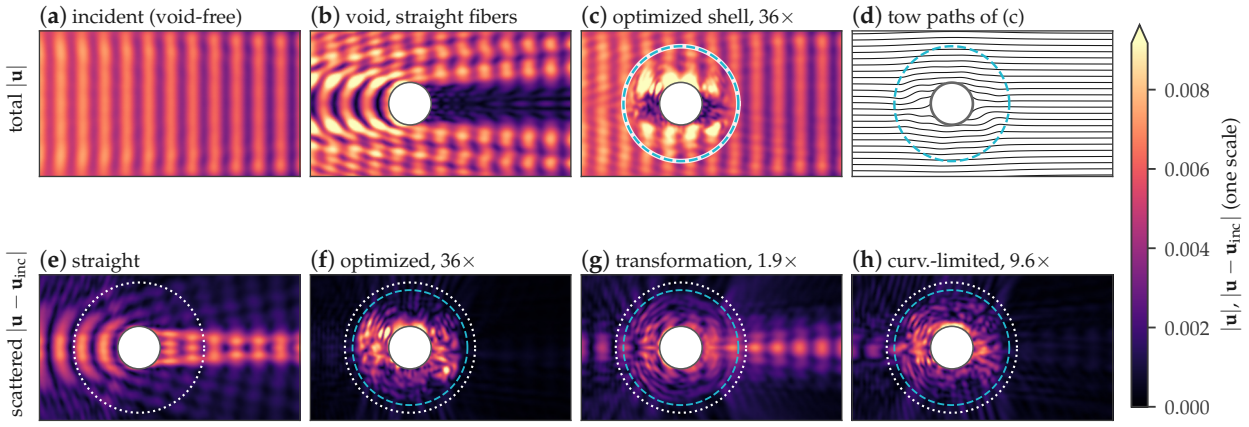


Figure 6: **The optimized shell cancels the scattered field outside the observation radius, not inside the shell.** Conforming traction-free void ($R_V = 0.55$) at $\omega = 50$, plate interior; all eight maps on one color scale, set by the maximum of the incident field (a) and clipped there. Top row, total field $|\mathbf{u}|$: **(a)** incident (void-free); **(b)** void with straight fibers: shadow and reflections; **(c)** the optimized shell (dashed circle: design shell R_{des}); **(d)** its tow paths. Bottom row, scattered field $|\mathbf{u} - \mathbf{u}_{\text{inc}}|$, with the far-field observation boundary R_C (dotted white): **(e)** straight fibers; **(f)** optimized; **(g)** transformation-cloak fiber directions on the fixed-contrast lamina; **(h)** optimized under the tow-curvature limit $\zeta_{\text{all}} = 1/R_V$. Reductions of the far-field mismatch are in the titles; they are confined to the design frequency (Table 6) and, for (c,f), to the design mesh.

Bounded-anisotropy cloaking is therefore real but tuned, to its frequency and to its mesh. Orientation fields that respect the tow-curvature limit cancel a scatterer's far field moderately and robustly; strong cancellation is narrowband and discretization-specific.

6.6 As built: tow spreading and defects

The designs are idealized fields with one angle per point, constant thickness and full coverage. Laying the lens with constant-width tows along its paths spreads and converges the tows at the rate $\nabla \cdot \boldsymbol{n}$, with $\boldsymbol{n} = (\cos\theta, \sin\theta)$. A course that starts fully covering its inflow edge reaches $\mathbf{x}$ with relative density $\tau = \exp(-\int \nabla \cdot \boldsymbol{n}\, \mathrm{d}\ell)$, integrated back along the path. For the lens, τ spans 0.003 to 52, so the field cannot be laid with uncut courses and needs tow cut/add. With cut/add

inside courses of N tows, the coverage mismatch averages half a tow at a course boundary. As in the defect-layer method [16], we model a defect area fraction $f_d = \frac{1}{2N}\min(1, |\ln\tau| / \ln(1 + 1/N))$ as resin gaps (stiffness and density of epoxy) or as overlaps (thickness $1 + f_d$; `wt_r1_lens.py asbuilt`). For $N = 4, 8$ and 16 the band-averaged redirection stays at 5.08–5.35×, against 5.2× for the idealized field, with the paper's layer. Wave redirection is insensitive to such defects because it depends on the fiber *direction*, which the defects do not change. The defect model, in particular its cap $f_d \leq 1/(2N)$, is an assumption, and tow width, process defects and multi-ply laminates are not resolved.

6.7 Design practice: a band-robust multifunctional panel

The preceding results imply two design rules. Optimize over a frequency band rather than at one frequency, so that the optimizer cannot buy performance with a resonance. Ask the fiber field to route energy, which channeling does broadband. A bolted 7×4 CFRP panel with four conforming fastener holes is designed this way. A single mirror-symmetric, curvature-feasible field ($\max|\zeta| = 5.4 \leq 1/R_{\text{tow}}$), optimized over 5 frequencies in $[44, 56]$, does four things at the band center (Fig. 7; parameters in Table 9):

- it routes the incident vibration onto a sensor patch (1.5×);
- it shields a payload core to 0.23× its straight-fiber energy;
- it keeps the four fatigue-critical fastener rims at 0.15×;
- it stiffens the bolted mount 1.18× at no added mass (stiffness is an explicit, weighted term of the objective).

Across the band, and why. Averaged over the design band $[44, 56]$ the sensor receives 1.56×, the payload 0.23× and the rims 0.27× their straight-fiber energies (Fig. 7f); the worst cases in the band are 1.05× (sensor), 0.32× (payload) and 0.49× (rims). Split as for the lens, the sensor's gain is redirection, not resonance: its share of the interior energy is 2.2–3.7× that with straight fibers across the band, while the whole interior field is *smaller*, 0.38–0.80× (Fig. 7e). A design optimized at the band center alone, with the same settings, routes slightly better but lets the rim protection degrade toward the lower band edge, to 1.09× the straight-fiber energy (band mean 0.52×). The band objective buys robust rim protection at a small cost in routing. The objective, a single signed region weight with logarithmic delivery and straight-fiber normalization, is given in Appendix E.

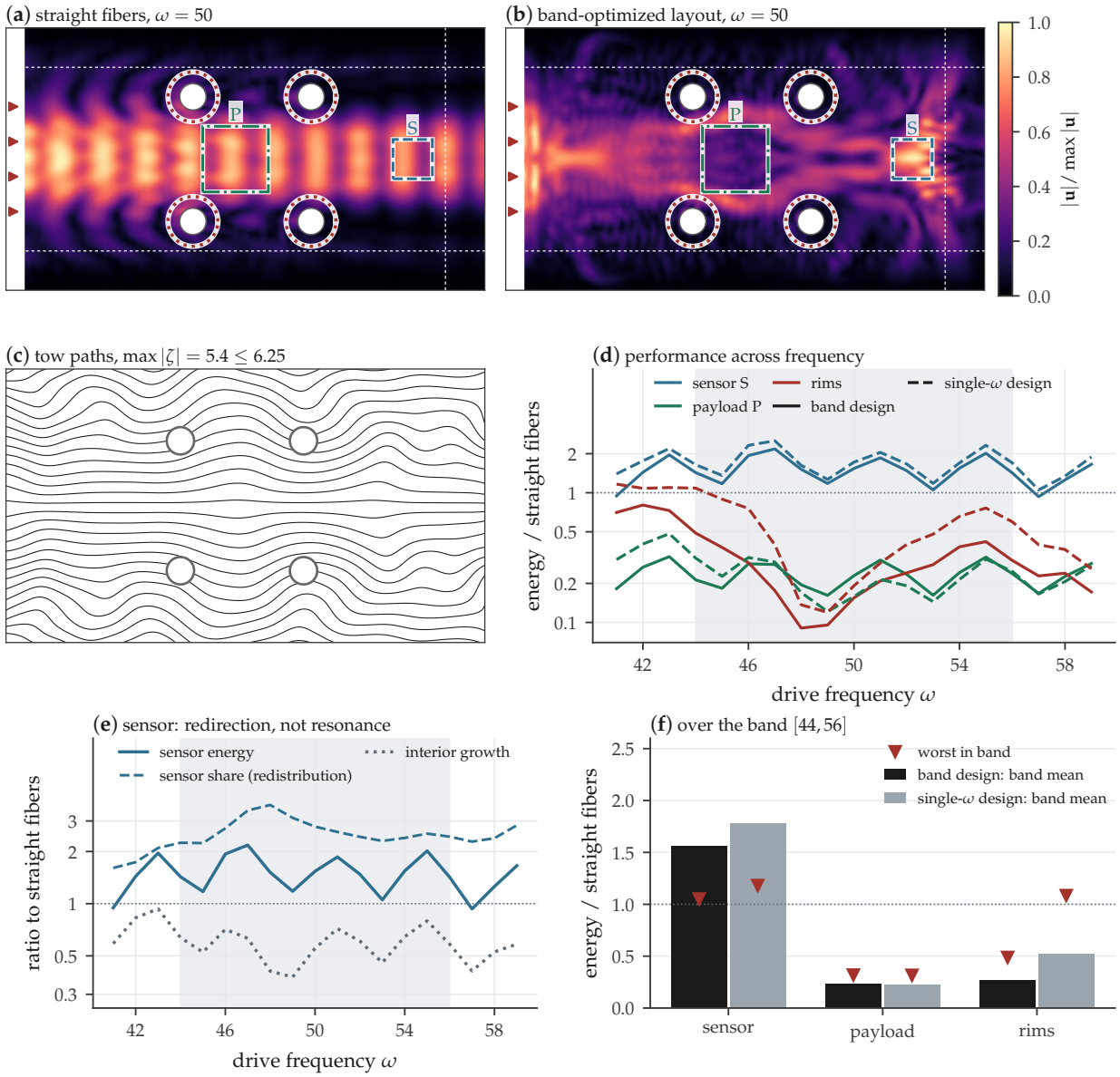


Figure 7: **One curvature-feasible fiber layout routes vibration to a sensor, shields a payload and protects fastener rims across a band, by redirection rather than resonance.** Four conforming fastener holes; Gaussian-beam drive along x at the left edge (arrows); dotted white: where the absorbing layer begins. **(a,b)** $|\mathbf{u}|$ at the band center with straight fibers and with the band-optimized layout, each normalized by its own maximum: the beam splits around the payload (P, green dash-dot) and reconverges on the sensor (S, blue dashed) while the fastener rims (red dotted) stay quiet. **(c)** Tow paths, curvature-feasible. **(d)** Energy in the sensor, payload and rims relative to straight fibers across frequency, band-optimized (solid) and optimized at $\omega = 50$ only (dashed); shaded: design band. **(e)** The sensor's energy ratio (solid) split into its share of the interior energy (dashed) and the whole-field growth (dotted). **(f)** Band means over $[44, 56]$ and worst cases in the band (triangles: lowest for the sensor, highest for payload and rims).

Table 7: Status of the predictions.

prediction	evidence	verdict
W1 isotropic plate: no control (identity; consistency check)	Remark 1; Table 4 ($s = 0$)	holds (exact)
W2 redirection is broadband, extra gain is resonant	Fig. 3: $G_\varphi \geq 3.7$ over $[37.5, 60]$; converged absorber 3.9–4.0; power share; growth peak $Q \approx 26, \propto 1/\eta$	holds over the tested wavelengths ($\lambda_\parallel \lesssim 1.5$); fails at long wavelength ($G_\varphi < 1$ at $\omega = 20, 25$)
W3 energy flux follows the fibers; rays predict the redirection	Fig. 4: 5.8° mean, unoptimized controls 1.0–4.2°; $G_{\text{ray}} = 4.2$–4.7	holds where fibers turn $\lesssim$ 4 rad per wavelength
W4 redirection grows with anisotropy	Table 4, Fig. 5a, best of four starts, two absorbing layers	holds for the best designs; finer shape within optimizer spread
W5 redirection unaffected while $R \lesssim 0.7\lambda_\parallel$; strong cloaking narrowband	Tables 5, 6; multi-start front; cloak mesh series	partly holds: gradual decline rather than failure at $R \sim \lambda$; cloak narrowband and mesh-tuned

7 Discussion

7.1 What is established, and how firmly

- **Mechanism.** Fiber orientation controls in-plane elastic waves by *channeling* their energy along the fibers. The evidence is fourfold: the flux follows the fibers within a few degrees wherever they turn by less than about 4 rad per wavelength; fields that were never optimized channel just as tightly; rays traced through the fiber field alone predict the redirection to within about 20%; and redirection vanishes for isotropic material and grows with anisotropy. Channeling is a qP, ray-regime effect: it fails at long wavelengths, near sharply converging fibers and at standing-wave nodes, and about a quarter to a half of the lens's kinetic energy is not qP-like.
- **Size.** With a converged absorbing layer the lens redirects 3.9–4.0× more energy into its target, band-averaged over the tested wavelengths, and absorbs several times more of the injected power there. The larger values obtained with a reflecting layer (5.2×) and at the design frequency (78× in G) include reflections and resonance respectively.
- **Limits.** The lever is bounded by the anisotropy and by the tow radius, with no loss while the minimum tow radius is below about 0.7 along-fiber wavelengths and a gradual decline beyond (best designs found, lower bounds).
- **Cancellation.** A full cloak must rely on interference, because exact cloaking needs anisotropy that no lamina has. Strong cancellation holds only near the design frequency and is tuned to the mesh it was optimized on; the curvature-limited design keeps a moderate 2.9–9.6× across $\omega \in [40, 60]$. The inextensible-fiber limit [21] and ground-plane cloaks [22, 23] escape this bound.

7.2 Consequences for how wave-control designs are reported

A finite, lightly damped plate driven at one frequency is a resonator, and an optimizer maximizing a field energy at that frequency exploits the resonance whenever it can. We therefore reaffirm three

practices, established for wave topology optimization [34, 33]: report redirection (a share of energy or of injected power) alongside, or instead of, raw gains; evaluate every design across a band and against a converged absorbing layer; and optimize band-averaged objectives when the application has a band (Sec. 6.7).

7.3 Relation to other orientation-design work

Orientation fields with smoothness or curvature limits have been designed for compliance [11, 13], frequency [6, 14, 15] and, together with topology and microstructure, for mechanical cloaking [12]. What differs here is the target (time-harmonic routing of in-plane elastic waves by a single lamina whose orientation is the only design field) and the analysis of energy transport, which identifies the mechanism and separates it from the resonance of the finite plate.

7.4 Limitations and scope

- **Model.** A single plane-stress lamina, with no through-thickness or flexural waves and constant thickness. The as-built check (Sec. 6.6) is a continuum estimate of tow spreading and defects; tow width, process defects and multi-ply laminates are not resolved, so the designs are tow-curvature-constrained, not shown to be manufacturable.
- **Absorbing layer.** The paper's layer reflects $|R| \approx 0.31$–0.34. We report every lens result against a converged layer as well; the redirection is about a quarter lower there, and the front and dose values, computed with the paper's layer, should be read with that factor. No perfectly matched layer was used.
- **Discretization.** Lens metrics change by at most about 7% under $2\times$ mesh refinement. The panel's sensor and payload ratios change by at most 6% on a finer mesh, its rim ratios by up to 0.10 in absolute terms. The optimized cloak is mesh-tuned, and its fixed-design refinement series has not converged.
- **Optimality.** The optimizations are local (MMA). The dose and front studies use multiple starts, and their values are best designs found, i.e. lower bounds on what is attainable.
- **Ray picture.** The ray prediction is single-pass and qP-only. It reproduces the size of the redirection but not its frequency dependence, which comes from reflections and interference.

7.5 Outlook

The channeling picture suggests where orientation design should go next: routing problems around prescribed holes and joints, where tows must wrap obstacles within the curvature limit; multi-objective layouts that trade wave routing against stiffness; combined topology and toolpath design, in which voids are cut rather than softened; and ray-based design, in which (8) replaces the wave solver in early design stages.

8 Conclusions

In a CFRP lamina the energy of quasi-longitudinal waves is skewed toward the fibers, by up to 65°, and 88% of wave normals carry their energy within 20° of them. A steered fiber field therefore acts as a field of energy streamlines wherever it turns slowly on the wavelength scale: the flux follows the fibers, and rays traced through the fibers predict the redirection. An orientation-only

Table 8: Complete parameter set.

quantity	lens	cloak
lamina $(E_{f1}, E_{f2}, G_{12}, \nu_{12})$, ρ	(131, 9, 5, 0.27), 1.6	
frequency ω (design)	40	50
structural damping η	0.02	0.02
domain	4×3	8×5, void $R_V = 0.55$ at $(3.2, 2.5)$
mesh	structured, 130×98 ($\times 2$ triangles)	`gmsh` conforming, $h = 0.045$ ($h/2$ on rim)
sponge (c_0, m), Eq. (16)	(80, 0.6)	(70, 0.6)
source (A, β, x_0), Eq. (17)	(30, 80, 0.1)	
CS-RBF (h_s, r)	(0.33, 0.7), mirror-symmetric	(0.28, 0.6), shell $R_{\text{des}} = 1.5$ (tapered)
objective region	box $(0.78L_x, L_y/2) \pm 0.18$	interior, $r > R_C = 1.7$
MMA move	0.10, 70 iterations	0.15 adaptive, 200–240 iterations
curvature limit ζ_{all}	none / front (Table 5)	none / $1/R_V = 1.82$
toolpath pitch d		0.16

lens redirects 3.9–4.0$\times$ more energy into its target over the tested band once the absorbing layer is converged, and routes several times more of the injected power there. Its much larger single-frequency gain is a resonance of the finite plate. The redirection vanishes for isotropic material, grows with anisotropy, and survives tow-curvature limits down to a minimum radius of about 0.7 along-fiber wavelengths, which sets the size and frequency at which a given placement process can use it. A full cloak cancels the far field strongly only at its design frequency and on its design mesh. Band-averaged design turns channeling into multifunctional layouts whose performance holds across the operating band.

Data and code availability. The solver, result files and the scripts that generate the reported results, tables and figures (Appendix F and the parameter tables) are in the repository `https://github.com/jtempelman95/wavetopo` (tag `R1`; see `docs/paper/README_REPRODUCE.md`).

Competing interests. The author declares no competing interests.

A Complete numerical parameters

Every constant below is the literal argument of the drivers (`examples/dolfinx/dolfinx_lens.py`, `dolfinx_cloak_conforming_vec.py`, `dolfinx_panel.py`) and the analysis scripts in `examples/analysis/`.

B Algorithm

Algorithm 1 Orientation design for wave control (`dolfinx`)

Require: mesh, material $\mathbf{Q}_f$, frequency ω, damping η, sponge $c(\mathbf{x})$, drive b, support sites $\{\mathbf{s}_i\}$, radius r, curvature limit ζ_{all}, void set $\mathcal{V}$

mesh: cut each void in $\mathcal{V}$ from the geometry (`gmsh`/OCC), refining the rim, so $\mathcal{T}_h$ conforms to it and the hole boundary is traction-free ▷ no soft void, no staircase

build $\mathbf{B}, \mathbf{B}_x, \mathbf{B}_y$ from (24), (27); build $\mathbf{S}$ from $\mathbf{S} = \frac{1}{2}(\mathbf{I} - \mathbf{\Pi})$

$\hat{\theta} \leftarrow \mathbf{0}$ ▷ straight-fiber baseline

solve U_{ref} on a *separate void-free* mesh and transfer it by non-matching interpolation (cloak only); record the baseline metric

for $\mu \in \{\mu_0, \dots, \mu_{\max}\}$ **do** ▷ penalty continuation

  for $t = 1, \dots, T$ **do**

    $\hat{\theta} \leftarrow \mathbf{S}\hat{\theta}$; $\quad \theta \leftarrow \mathbf{B}\hat{\theta}$ ▷ symmetrize, then interpolate

    assemble $\mathcal{A}(\boldsymbol{\theta})$ as the real 4×4 block (13) ▷ voids already absent from $\mathcal{T}_h$

    forward: factor $\mathcal{A} = \mathbf{LU}$, solve $\mathcal{A}\mathsf{U} = \mathsf{b}$

    evaluate J, and E (focus energy) or the far-field mismatch

    adjoint: solve $\mathcal{A}^{\mathsf{T}}\mathbf{\Lambda} = -\partial J/\partial \mathsf{U}$ (sparse LU of $\mathcal{A}^{\mathsf{T}}$)

    $d_\ell \leftarrow \partial R/\partial\theta_\ell$ by UFL differentiation (23)

    $\boldsymbol{\zeta} \leftarrow \cos\theta\,(\mathbf{B}_x\hat{\theta}) + \sin\theta\,(\mathbf{B}_y\hat{\theta})$; $\quad \mathcal{C} \leftarrow \langle \max(0, \zeta^2 - \zeta_{\text{all}}^2) \rangle$

    $\mathbf{g} \leftarrow$ (23) with the curvature penalty of §5.5 ▷ $\mathbf{S}^{\mathsf{T}}\mathbf{B}^{\mathsf{T}}$ plus curvature term

    $\hat{\theta} \leftarrow \mathrm{MMA}\big(\hat{\theta}, \mathbf{g};\ -\frac{\pi}{2} \le \hat{\theta} \le \frac{\pi}{2}\big)$

  end for

  if $\max_c |\zeta_c| \le \zeta_{\text{all}}$ **then break** ▷ feasible

  end if

end for

delineate toolpaths: sign-consistent branch of $(\cos 2\theta, \sin 2\theta)$; solve $\nabla\Phi = \mathbf{g}$ by `lsqr` (25); tows $\chi = \frac{1}{2} + \frac{1}{2}\cos(2\pi\Phi/d)$, masked off $\mathcal{V}$

return $\hat{\theta}$, θ, U, gain/reduction, χ

C Detailed derivations (modeling)

This appendix carries every modeling step in full, so each equation of §2–§4.3 can be re-derived and independently checked.

C.1 Orthotropic plane-stress stiffness $\mathbf{Q}_f$

In the fiber (material) frame $(1, 2)$, under plane stress ($\sigma_{33} = \sigma_{13} = \sigma_{23} = 0$), the orthotropic compliance relating the engineering strain $\boldsymbol{\varepsilon} = [\varepsilon_{11}, \varepsilon_{22}, \gamma_{12}]^{\mathsf{T}}$ to the stress $\boldsymbol{\sigma} = [\sigma_{11}, \sigma_{22}, \sigma_{12}]^{\mathsf{T}}$ is

$$\boldsymbol{\varepsilon} = \mathbf{S}\,\boldsymbol{\sigma}, \qquad \mathbf{S} = \begin{bmatrix} 1/E_{f1} & -\nu_{21}/E_{f2} & 0 \\ -\nu_{12}/E_{f1} & 1/E_{f2} & 0 \\ 0 & 0 & 1/G_{12} \end{bmatrix}, \tag{29}$$

with the reciprocity $\nu_{12}/E_{f1} = \nu_{21}/E_{f2}$, i.e. $\nu_{21} = \nu_{12}E_{f2}/E_{f1}$, which makes $\mathbf{S}$ symmetric. The stiffness is $\mathbf{Q}_f = \mathbf{S}^{-1}$. The upper 2×2 block inverts with determinant $\Delta = \frac{1}{E_{f1}E_{f2}}(1 - \nu_{12}\nu_{21})$,

giving

$$Q_{11} = \frac{E_{f1}}{1-\nu_{12}\nu_{21}}, \quad Q_{22} = \frac{E_{f2}}{1-\nu_{12}\nu_{21}}, \quad Q_{12} = \frac{\nu_{12}E_{f2}}{1-\nu_{12}\nu_{21}}, \quad Q_{66} = G_{12}, \tag{30}$$

the entries of (2) (`Qf_matrix`). The shear row decouples, hence $Q_{16} = Q_{26} = 0$.

C.2 Rotation to the global frame and $\partial \mathbf{D}_f / \partial\theta$

Let θ be the fiber angle and $(c, s) = (\cos\theta, \sin\theta)$. Rotating the material-frame engineering strain to the global frame is $\boldsymbol{\varepsilon}_{\text{mat}} = \mathbf{T}(\theta)^{\mathsf{T}}\boldsymbol{\varepsilon}$ with the engineering-strain rotation

$$\mathbf{T}(\theta) = \begin{bmatrix} c^2 & s^2 & -2sc \\ s^2 & c^2 & 2sc \\ sc & -sc & c^2 - s^2 \end{bmatrix}, \tag{31}$$

so that the global stiffness is a congruence transform of the material one,

$$\mathbf{D}_f(\theta) = \mathbf{T}(\theta)\,\mathbf{Q}_f\,\mathbf{T}(\theta)^{\mathsf{T}}. \tag{32}$$

Differentiating entry-by-entry ($c' = -s$, $s' = c$),

$$\mathbf{T}'(\theta) = \frac{\mathrm{d}\mathbf{T}}{\mathrm{d}\theta} = \begin{bmatrix} -2sc & 2sc & -2(c^2 - s^2) \\ 2sc & -2sc & 2(c^2 - s^2) \\ c^2 - s^2 & -(c^2 - s^2) & -4sc \end{bmatrix}, \tag{33}$$

and by the product rule on (32),

$$\frac{\partial \mathbf{D}_f}{\partial\theta} = \mathbf{T}'\mathbf{Q}_f\mathbf{T}^{\mathsf{T}} + \mathbf{T}\mathbf{Q}_f{\mathbf{T}'}^{\mathsf{T}}, \tag{34}$$

which is the object every orientation sensitivity is built from (cf. (23)).

C.3 Exact five-term Fourier decomposition of $\mathbf{D}_f(\theta)$

The `dolfinx` implementation behind every result in this paper evaluates (32)–(34) directly, as a congruence product differentiated symbolically by UFL. The rotation nevertheless has a sharper structure, which we derive here because it exposes the material invariants through which every orientation effect acts (Sec. 3) and because it gives an equivalent closed form, cheaper to evaluate, that a structured-grid implementation can use: $\mathbf{D}_f(\theta)$ *is a trigonometric polynomial of degree four in* θ, *exactly*.

Start from the entries of $\mathbf{T}(\theta)$. Each is a quadratic monomial in (c, s) — one of c^2, s^2, or sc — and the double-angle identities write every such monomial in the fixed basis $\{1, \cos 2\theta, \sin 2\theta\}$:

$$c^2 = \tfrac{1}{2}(1 + \cos 2\theta), \qquad s^2 = \tfrac{1}{2}(1 - \cos 2\theta), \qquad sc = \tfrac{1}{2}\sin 2\theta. \tag{35}$$

So every entry of $\mathbf{T}$ — and hence of $\mathbf{T}^{\mathsf{T}}$ — lives in $\operatorname{span}\{1, \cos 2\theta, \sin 2\theta\}$. The congruence (32) multiplies two $\mathbf{T}$-factors (with the constant $\mathbf{Q}_f$ between them), so each entry of $\mathbf{D}_f$ is a sum of *products of two* such first-harmonic terms. The product-to-sum identities collapse those products to the second harmonic and no higher,

$$\cos 2\theta \cos 2\theta = \tfrac{1}{2}(1 + \cos 4\theta), \quad \sin 2\theta \sin 2\theta = \tfrac{1}{2}(1 - \cos 4\theta), \quad \cos 2\theta \sin 2\theta = \tfrac{1}{2}\sin 4\theta, \tag{36}$$

so nothing beyond 4θ can appear. Every entry of $\mathbf{D}_f(\theta)$ therefore lies in the five-dimensional space $\operatorname{span}\{1, \cos 2\theta, \sin 2\theta, \cos 4\theta, \sin 4\theta\}$, and collecting the coefficient of each basis function into its own constant 3×3 matrix gives the *exact* decomposition

$$\boxed{\mathbf{D}_f(\theta) = \sum_{s=0}^{4} c_s(\theta)\, \mathbf{C}_s}, \qquad \mathbf{c}(\theta) = \big[\,1,\ \cos 2\theta,\ \sin 2\theta,\ \cos 4\theta,\ \sin 4\theta\,\big], \tag{37}$$

in which the five $\mathbf{C}_s$ depend only on $\mathbf{Q}_f$ (each symmetric, since $\mathbf{D}_f$ is). They are recovered once, offline, by projecting $\mathbf{D}_f(\theta)$ onto the basis $\mathbf{c}$ over a dense θ-sample — an exact fit, not an approximation, because the five functions span the range; in the implementation the residual of the least-squares recovery is 8.5×10^{-16}, i.e. machine zero, which is the numerical confirmation that the degree is really four and not higher.

Two consequences make this the preferred form. First, the derivative no longer needs the product rule (34); it is just the (elementwise) derivative of the scalar basis,

$$\frac{\partial \mathbf{D}_f}{\partial \theta} = \sum_{s=0}^{4} c_s'(\theta)\, \mathbf{C}_s, \qquad \mathbf{c}'(\theta) = \big[\,0,\ -2\sin 2\theta,\ 2\cos 2\theta,\ -4\sin 4\theta,\ 4\cos 4\theta\,\big], \tag{38}$$

which is exact to round-off and shares the constant $\mathbf{C}_s$ with the value (37), so value and gradient cost one small matrix combination each. Second, the decomposition passes *through* the element integral. The fiber element stiffness is $\mathbf{k}_f(\theta) = \int_e \mathbf{B}^{\mathsf{T}} \mathbf{D}_f(\theta)\, \mathbf{B}$, and because the $c_s(\theta)$ are scalars they factor out of the integral,

$$\mathbf{k}_f(\theta) = \sum_{s=0}^{4} c_s(\theta) \underbrace{\int_e \mathbf{B}^{\mathsf{T}} \mathbf{C}_s\, \mathbf{B}}_{\mathbf{k}_f^{(s)}} = \sum_{s=0}^{4} c_s(\theta)\, \mathbf{k}_f^{(s)}, \tag{39}$$

so an implementation can cache five fixed matrices $\mathbf{k}_f^{(s)}$ per element (for a linear triangle, $\mathbf{k}_f^{(s)} = A\, \mathbf{B}^{\mathsf{T}} \mathbf{C}_s \mathbf{B}$, since $\mathbf{B}$ is constant), and every value or θ-derivative of the element stiffness is a five-term weighted sum of them. The two forms give identical stiffness and gradients; the symbolic form is what we use.

C.4 Isotropy annihilation (proof of Remark 1)

$\mathbf{Q}_f$ is isotropic iff $Q_{11} = Q_{22}$ and $Q_{66} = \frac{1}{2}(Q_{11} - Q_{12})$. Then $\mathbf{Q}_f = \lambda \mathbf{1}\mathbf{1}^{\mathsf{T}} + 2\mu\, \mathbf{P}$ with $\mathbf{1} = [1, 1, 0]^{\mathsf{T}}$ and $\mathbf{P}$ the deviatoric projector, both invariant under the in-plane rotation $\mathbf{T}(\theta)$ (an isotropic tensor is unchanged by rotation): $\mathbf{T}\mathbf{Q}_f\mathbf{T}^{\mathsf{T}} = \mathbf{Q}_f$ for all θ. Hence $\mathbf{D}_f(\theta) \equiv \mathbf{Q}_f$ and, by (34), $\partial \mathbf{D}_f / \partial \theta \equiv \mathbf{0}$; by (23) every orientation sensitivity vanishes. Orientation is a dead variable exactly when the material is isotropic. □

C.5 Elastodynamics, time-harmonic reduction, damping, sponge

Momentum balance for the displacement $\mathbf{u}(\mathbf{x}, t)$ with mass density ρ and the plane-stress constitutive law $\boldsymbol{\sigma} = \mathbf{D}_f(\theta)\boldsymbol{\varepsilon}(\mathbf{u})$ is $\nabla \cdot \boldsymbol{\sigma} = \rho\, \ddot{\mathbf{u}}$. Seeking time-harmonic motion $\mathbf{u}(\mathbf{x}, t) = \operatorname{Re}\{\mathbf{u}(\mathbf{x}) e^{+\mathrm{i}\omega t}\}$ turns $\ddot{\mathbf{u}} \to -\omega^2 \mathbf{u}$ and $\dot{\mathbf{u}} \to \mathrm{i}\omega \mathbf{u}$. Structural (hysteretic) damping multiplies the stiffness by $(1 + \mathrm{i}\eta)$, and a mass-proportional absorbing sponge $c(\mathbf{x}) \geq 0$ adds a term $+\mathrm{i}\omega c\, \mathbf{u}$ (the $\dot{\mathbf{u}} \to \mathrm{i}\omega \mathbf{u}$ of a viscous force $c\dot{\mathbf{u}}$), giving the strong form (4). Multiplying by a test function $\mathbf{v}$ and integrating by

parts (the sponge edges carry the outgoing radiation, the driven edge the source) yields the weak form and, after P_1 interpolation, the complex system (12) with $\mathbf{K} = \int \boldsymbol{\varepsilon}(\cdot)^{\mathsf{T}}\mathbf{D}_f\boldsymbol{\varepsilon}(\cdot)$, $\mathbf{M} = \int \rho(\cdot)\cdot(\cdot)$, $\mathbf{C} = \int c(\cdot)\cdot(\cdot)$ (`_forms`).

C.6 Exact real four-field split

Write the complex operator of (12) as $\mathbf{D} = (1+\mathrm{i}\eta)\mathbf{K} - \omega^2\mathbf{M} + \mathrm{i}\omega\mathbf{C} = A + \mathrm{i}B$ with

$$A = \mathbf{K} - \omega^2\mathbf{M}, \qquad B = \eta\mathbf{K} + \omega\mathbf{C}, \tag{40}$$

both real symmetric. Splitting $\mathbf{u} = \mathbf{u}_r + \mathrm{i}\mathbf{u}_i$ and $\mathbf{f} = \mathbf{f}_r + \mathrm{i}\mathbf{f}_i$, the equation $(A+\mathrm{i}B)(\mathbf{u}_r + \mathrm{i}\mathbf{u}_i) = \mathbf{f}_r + \mathrm{i}\mathbf{f}_i$ separates into real and imaginary parts,

$$A\mathbf{u}_r - B\mathbf{u}_i = \mathbf{f}_r, \qquad B\mathbf{u}_r + A\mathbf{u}_i = \mathbf{f}_i, \tag{41}$$

i.e. the real block system (13) on $\mathsf{U} = (u_{xr}, u_{yr}, u_{xi}, u_{yi})$. It is exactly $[\mathrm{Re}, -\mathrm{Im}; \mathrm{Im}, \mathrm{Re}]$ of $\mathbf{D}$, so a real `PETSc` build solves the complex problem without complex scalars.

D Detailed derivations (optimization)

D.1 Discrete adjoint and design sensitivity

Let the assembled forward problem be $\mathcal{A}(\boldsymbol{\theta})\mathsf{U} = \mathsf{b}$ with $\mathcal{A} = [A, -B; B, A]$, and let $J(\mathsf{U})$ be an objective. Form the Lagrangian with a multiplier $\boldsymbol{\Lambda}$, $\mathcal{L} = J(\mathsf{U}) + \boldsymbol{\Lambda}^{\mathsf{T}}(\mathcal{A}\mathsf{U} - \mathsf{b})$; on the constraint manifold $\mathcal{L} = J$. Its total derivative in a design θ_ℓ is

$$\frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \frac{\partial J}{\partial \mathsf{U}}^{\mathsf{T}}\frac{\mathrm{d}\mathsf{U}}{\mathrm{d}\theta_\ell} + \boldsymbol{\Lambda}^{\mathsf{T}}\Big(\mathcal{A}\frac{\mathrm{d}\mathsf{U}}{\mathrm{d}\theta_\ell} + \frac{\partial \mathcal{A}}{\partial \theta_\ell}\mathsf{U}\Big) = \Big(\frac{\partial J}{\partial \mathsf{U}} + \mathcal{A}^{\mathsf{T}}\boldsymbol{\Lambda}\Big)^{\mathsf{T}}\frac{\mathrm{d}\mathsf{U}}{\mathrm{d}\theta_\ell} + \boldsymbol{\Lambda}^{\mathsf{T}}\frac{\partial \mathcal{A}}{\partial \theta_\ell}\mathsf{U}. \tag{42}$$

Choosing $\boldsymbol{\Lambda}$ to annihilate the bracket gives the adjoint equation and reduces the sensitivity to the explicit design dependence of the residual $R(\boldsymbol{\theta}) = \boldsymbol{\Lambda}^{\mathsf{T}}\big(\mathcal{A}(\boldsymbol{\theta})\mathsf{U} - \mathsf{b}\big)$ (with $\mathsf{U}, \boldsymbol{\Lambda}$ frozen):

$$\mathcal{A}^{\mathsf{T}}\boldsymbol{\Lambda} = -\frac{\partial J}{\partial \mathsf{U}}, \qquad \frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \boldsymbol{\Lambda}^{\mathsf{T}}\frac{\partial \mathcal{A}}{\partial \theta_\ell}\mathsf{U}. \tag{43}$$

Only $\mathbf{K}(\boldsymbol{\theta})$ (hence A, B) depends on θ, and only through $\partial\mathbf{D}_f/\partial\theta$ (34); expanding $\boldsymbol{\Lambda} = (\boldsymbol{\lambda}_r, \boldsymbol{\lambda}_i)$ block-wise and using $\partial A/\partial\theta_\ell = \partial\mathbf{K}/\partial\theta_\ell$, $\partial B/\partial\theta_\ell = \eta\,\partial\mathbf{K}/\partial\theta_\ell$ gives the explicit cell integral (23). The mass, sponge and source terms carry no θ and drop out, so the gradient is entirely built from $\partial\mathbf{D}_f/\partial\theta$ — which vanishes for an isotropic material, making Remark 1 a statement about the gradient itself. For a quadratic objective $J = \int w|\mathbf{u}|^2$ the adjoint load is linear, $\partial J/\partial\mathsf{U} = 2\,\mathrm{diag}(w)\mathsf{U}$ (a signed weight w gives the multi-region objectives of Sec. 6.7); in the code both the adjoint load and the residual derivative are single `ufl.derivative` calls (`_sensitivity`), FD-verified to $\sim 10^{-6}$.

D.2 CS-RBF chain rule and its analytic derivatives

With $\theta = B\hat{\theta}$ (24), the reduced gradient is $\partial J/\partial\hat{\theta} = B^{\mathsf{T}}(\partial J/\partial\theta)$. For the curvature constraint we also need $\partial_x\theta, \partial_y\theta$. Writing $W_{ci} = \phi(q_{ci})$ and $S_c = \sum_j W_{cj}$ so $B_{ci} = W_{ci}/S_c$, the chain rule on $q_{ci} = \|\mathbf{x}_c - \mathbf{s}_i\|/r$ with $\phi'(q) = -20q(1-q)^3$ gives $\partial_x W_{ci} = \phi'(q_{ci})\,\partial_x q_{ci} = -\frac{20(1-q_{ci})^3}{r^2}(x_c - s_{ix})$, and the quotient rule on B_{ci} yields (27), i.e. $\mathbf{B}_x = S^{-1}W_x - \mathrm{diag}(S^{-1}W_x\mathbf{1})\,B$ (and $\mathbf{B}_y$ likewise); the second term is the derivative of the partition-of-unity normalization, without which $\sum_i(\mathbf{B}_x)_{ci} \neq 0$ and the curvature would be biased (`csrbf_operators`).

D.3 Fiber-curvature functional and its Jacobian

The fiber-path curvature is the directional derivative of θ along the fiber $\mathbf{n} = (\cos\theta, \sin\theta)$, $\zeta = \cos\theta\,\theta_x + \sin\theta\,\theta_y$ with $\theta_x = \mathbf{B}_x\hat{\theta}$, $\theta_y = \mathbf{B}_y\hat{\theta}$. Differentiating w.r.t. $\hat{\theta}$ (product rule, $\partial\theta/\partial\hat{\theta} = B$),

$$\frac{\partial\zeta}{\partial\hat{\theta}} = \operatorname{diag}(-\sin\theta\,\theta_x + \cos\theta\,\theta_y)\,B + \operatorname{diag}(\cos\theta)\,\mathbf{B}_x + \operatorname{diag}(\sin\theta)\,\mathbf{B}_y. \tag{44}$$

The penalty $\mathcal{C} = \langle \max(0, \zeta^2 - \zeta_{\text{all}}^2)\rangle$ has gradient $\partial\mathcal{C}/\partial\hat{\theta} = \frac{2}{n}(\partial\zeta/\partial\hat{\theta})^{\mathsf{T}}[\zeta \odot \mathbf{1}_{\{\zeta^2 > \zeta_{\text{all}}^2\}}]$ (restricted to solid cells for the cloak), FD-verified to $\sim 10^{-10}$ (`curl_penalty`). It is added to the objective with a continuation weight μ ramped until the constraint is met.

D.4 Phase-field toolpath extraction

The tow field is the level set of Φ with $\nabla\Phi = \mathbf{g} = [-\sin\theta, \cos\theta]^{\mathsf{T}}$ (unit, orthogonal to the fiber). This overdetermined system (two equations, one unknown) is solved in the least-squares sense $\min_\Phi \int \|\nabla\Phi - \mathbf{g}\|^2$. Its Euler–Lagrange condition is the Poisson equation $\nabla^2\Phi = \nabla\cdot\mathbf{g}$ with the *natural* Neumann data $\partial\Phi/\partial n = \mathbf{g}\cdot\mathbf{n}$ on $\partial\Omega$. The boundary flux is essential: dropping it (homogeneous Neumann) leaves only $\nabla^2\Phi = \nabla\cdot\mathbf{g}$, whose solution for a nearly divergence-free $\mathbf{g}$ collapses to $\Phi \approx$ const ($|\nabla\Phi| \to 0$), i.e. washed-out tows. We therefore assemble the discrete forward-difference gradient operator $G = [G_x; G_y]$ on the grid and solve $\min_\Phi \|G\Phi - \mathbf{g}\|^2$ directly (`lsqr`, one pinned node fixing the constant), which enforces $|\nabla\Phi| \approx 1$, and hence a constant tow pitch, where the director field is nearly divergence-free. The tows are the peaks of $\chi = \frac{1}{2} + \frac{1}{2}\cos(2\pi\Phi/d)$ (`fiber_phase`). A seam-free director branch is obtained beforehand by region-growing sign propagation (`_orient_consistent`).

D.5 Symmetry projector

The map $\mathbf{\Pi}$ exchanging each support site with its mirror image is an involutive permutation ($\mathbf{\Pi}^2 = \mathbf{I}$, $\mathbf{\Pi}^{\mathsf{T}} = \mathbf{\Pi}$), so with $\mathbf{S} = \frac{1}{2}(\mathbf{I} - \mathbf{\Pi})$, $\mathbf{S}^{\mathsf{T}} = \mathbf{S}$ and $\mathbf{S}^2 = \frac{1}{4}(\mathbf{I} - 2\mathbf{\Pi} + \mathbf{\Pi}^2) = \frac{1}{2}(\mathbf{I} - \mathbf{\Pi}) = \mathbf{S}$. Thus $\mathbf{S}$ is the *orthogonal* projector onto $\{\hat{\theta} : \mathbf{\Pi}\hat{\theta} = -\hat{\theta}\}$, the mirror-symmetric fiber fields. Being self-adjoint, the same operator symmetrizes the design and the gradient, so imposing symmetry introduces no inconsistency between the two — the projected gradient is exactly the gradient of the restricted problem. □

D.6 MMA subproblem

Each iteration the Method of Moving Asymptotes [40] replaces $J, \mathcal{C}$ by convex separable approximations built from moving asymptotes $L_\ell < \hat{\theta}_\ell < U_\ell$,

$$\tilde{J}(\hat{\theta}) = r + \sum_\ell \Big(\frac{p_\ell}{U_\ell - \hat{\theta}_\ell} + \frac{q_\ell}{\hat{\theta}_\ell - L_\ell}\Big), \quad \begin{cases} p_\ell = (U_\ell - \hat{\theta}_\ell^{(k)})^2\,(\partial_\ell J)^+ \\ q_\ell = (\hat{\theta}_\ell^{(k)} - L_\ell)^2\,(\partial_\ell J)^- \end{cases} \tag{45}$$

($(\cdot)^\pm$ the positive/negative parts), with the asymptotes contracted or expanded by the sign history of $\hat{\theta}_\ell$ and a move limit per step. The separable convex subproblem is solved by its (cheap) dual. Because the raw objectives are $O(10^{-4})$ we feed MMA the scale-invariant log-gradient $g_\ell = (\partial_{\theta_\ell} J)/J$ for the lens focus, restoring $O(1)$ sensitivities; the cloak uses the raw gradient (well scaled). Bounds are $\hat{\theta}_\ell \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ (`cfrp_optimizer.py : MMA`).

Table 9: Parameters of the multifunctional panel. Band design: `dolfinx_panel.py --band 44,47,50,53,56 --lam-p 3 --stiff 0.3`; the single-frequency design uses `--band 50` with the same settings.

plate, lamina, damping	7×4; lamina of Table 8; $\eta = 0.02$
mesh	`gmsh` conforming, $h = 0.055$ ($h/2$ on hole rims)
sponge (c_0, m)	$(80,\ 0.6)$ on the right, top and bottom edges
drive	$55\, e^{-80(x-0.15)^2} e^{-((y-2)/0.95)^2}$ (x-directed beam)
fastener holes	radius 0.20 at $(2.55, 1.05)$, $(2.55, 2.95)$, $(4.35, 1.05)$, $(4.35, 2.95)$
rim regions F	annuli of width 0.18 around the holes
payload P	square $(3.2, 2.0) \pm 0.5$
sensor S	square $(5.9, 2.0) \pm 0.3$
weights $(W_S, \lambda_P, \lambda_F, \lambda_O)$	$(1,\ 3,\ 1,\ 0.5)$ (strong shield)
stiffness term	clamped at the four bolt rims, unit downward load over P; $w_c = 0.3$
band	$\omega \in \{44, 47, 50, 53, 56\}$ (single-frequency design: $\omega = 50$)
tow limit	$R = 0.16$ ($\zeta_{\text{all}} = 6.25$); achieved $\max \lvert\zeta\rvert = 5.4$
optimizer	MMA, 110 iterations from straight fibers
CS-RBF	mirror-symmetric support grid, as for the lens

E Multifunctional panel objective

With the region energies $E_R = \int_{\Omega_R} |\mathbf{u}|^2 \, \mathrm{d}\mathbf{x}$, the four wave jobs compose into a single scalar

$$\mathcal{J}_{\text{wave}} = W_S \log \frac{E_S}{E_S^0} - \lambda_P \frac{E_P}{E_P^0} - \lambda_F \frac{E_F}{E_F^0} - \lambda_O \frac{E_O}{E_O^0}, \tag{46}$$

with the regions made *disjoint* by priority (rims > payload > sensor > outside) so no cell is rewarded and penalized at once, and each term divided by its straight-fiber baseline E_R^0 so all four are $O(1)$ and directly comparable. Two choices matter. The delivery reward is *logarithmic*: with a plain ratio a bright sensor term would swamp the rim penalty and the optimizer would buy delivery by letting the fasteners get busier, whereas log has diminishing returns so protecting the rims stays worthwhile however bright the port already is. The confinement term penalizes interior energy *outside* the target regions, which forces a clean routed beam rather than a diffraction spray. The gradient still needs one adjoint solve per state: for $\mathcal{J}_{\text{wave}} = \sum_R c_R E_R$ (the log term contributing $c_S = W_S / E_S$, re-evaluated each iterate) the adjoint load is the summed region weight $w = \sum_R (\partial \mathcal{J} / \partial E_R)\, \mathbf{1}_{\Omega_R}$, and the discrete-adjoint sensitivity of §5 returns $\mathrm{d}\mathcal{J}_{\text{wave}}/\mathrm{d}\theta_\ell$; the CS-RBF transpose then gives the reduced gradient. The panel is mirror-symmetric about $y = L_y/2$, enforced exactly by a symmetric support grid and the antisymmetric projector of §5.4.

Operational bandwidth. A single-frequency optimum is a single-tone trick. We make the design robust across an operational band $[44, 56]$ by averaging the objective over 5 sample frequencies,

$$\mathcal{J} = \frac{1}{K} \sum_{k=1}^{K} \mathcal{J}_{\text{wave}}(\omega_k; \theta), \qquad \frac{\mathrm{d}\mathcal{J}}{\mathrm{d}\theta_\ell} = \frac{1}{K} \sum_k \frac{\mathrm{d}\mathcal{J}_{\text{wave}}(\omega_k)}{\mathrm{d}\theta_\ell}, \tag{47}$$

one forward and one adjoint solve per frequency per iterate. The band optimization earns its keep at the band *edges*: a design tuned at the center alone holds the shield and delivery across the band but lets the fastener protection degrade toward the edges, which the averaged design restores (Figure 7d).

Table 10: Software environment and solver settings.

FE library	`dolfinx` 0.7.3, `ufl` 2023.2.0, `basix` 0.7.0
linear algebra	PETSc 3.20.6, real scalars (float64); MUMPS sparse direct LU (forward and adjoint)
meshes	structured triangles (lens, $n_x = 130$); `gmsh` 4.15 OCC, conforming holes, rim refinement $h/2$
elements	$[\mathbb{P}_1]^4$ state (real four-field split), $\mathbb{DG}_0$ orientation
optimizer	MMA [40], move limit 0.10 (lens), 0.15 (cloak); iterations in Table 8
other	NumPy 1.26.4, SciPy 1.15.3; figures with Matplotlib (pgf backend)

Stiffness. The same orientation is scored for static stiffness. The panel is clamped at the four bolt rims (the bolted mount) and carries a service load over the payload footprint; the compliance $c = \mathbf{f}^\top \mathbf{u}$ and its self-adjoint orientation sensitivity $\mathrm{d}c/\mathrm{d}\theta_\ell = -\int_{\Omega_\ell} \boldsymbol{\varepsilon}(\mathbf{u})^\top (\partial \mathbf{D}_f/\partial\theta)\boldsymbol{\varepsilon}(\mathbf{u})\,\mathrm{d}\mathbf{x}$ — one extra linear solve, reusing the static factorization — enter the objective as $-w_c\, c/c_0$. The fiber angle is largely spent on the wave, so with this weight ($w_c = 0.3$, an explicit term of the objective) the band-optimized layout reaches $1.18\times$ the straight-fiber stiffness at no added mass; pushing stiffness harder trades directly against the routing and shielding. A density/topology field would decouple stiffness and mass from the wave, but a SIMP soft-void contradicts the conforming-void modeling used throughout and is not pursued here.

F Numerical verification and reproducibility

Software, solver and files. Table 10 lists the software and solver settings. Every linear solve is a sparse direct LU factorization (MUMPS through PETSc), so there is no iterative tolerance; the forward and adjoint residuals are at round-off, and the discrete power balance (20) closes to 10^{-13}. Every quoted number is a macro generated from the result files by `wt_paper_assets.py` (`docs/paper/wt_numbers.tex`), and every design shown is stored with its mesh centroids and orientation in `results/` (e.g. `dolfinx_lens_data.npz`, `lens_absorber.npz`, `cloak_headtohead.npz`, `dolfinx_panel_band_repro_data.npz`).

Checks. The checks are produced by the scripts `wt_verify.py`, `cloak_fine.py`, `lens_absorber.py`, `lens_flux_control.py` and, for this revision, `wt_r1_lens.py` (power balance, flux map, fine sweep, absorber series, as-built), `wt_r1_rays.py`, `wt_r1_cloak.py` (mesh series, wider layer, scattered fields), `wt_r1_dose_ms.py` and `wt_r1_panel.py`, all in `examples/analysis/`. Table 11 lists the lens checks.

- **Mesh.** The lens is re-solved with the optimized field resampled (through $(\cos 2\theta, \sin 2\theta)$, the field being a smooth CS-RBF field) on structured meshes with $n_x = 130$ (the design mesh: about 9 P1 elements per wavelength of the slowest, quasi-shear wave at $\omega = 40$, and about 46 per along-fiber qP wavelength), 195 and 260. At $\omega = 40$, G stays within 73–78 and G_φ within 5.72–5.96. The design's sharpest orientation gradient length, $L_\theta = 0.15$, spans about 5 elements of the design mesh. The cloak's fixed-design refinement series ($h = 0.045$ to 0.025) is discussed in Sec. 6.5.
- **Damping.** At $\omega = 40$ the redirection is independent of the damping ($G_\varphi = 5.8, 5.8, 5.8$ for $\eta = 0.01, 0.02, 0.04$), while the interior growth falls roughly as $1/\eta$ (23.3, 13.4, 6.6: a ratio of 1.7–2.0 per doubling of η), and so does the gain (135, 78, 38). This directly confirms that the gain beyond G_φ is resonant.

Table 11: Numerical verification of the lens (fixed optimized design).

check	setting	G	G_φ	growth
lens mesh, $\omega = 40$	$n_x = 130$	78.0	5.82	13.41
lens mesh, $\omega = 40$	$n_x = 195$	73.2	5.72	12.80
lens mesh, $\omega = 40$	$n_x = 260$	74.4	5.96	12.48
lens mesh, $\omega = 50$	$n_x = 130$	5.5	6.01	0.91
lens mesh, $\omega = 50$	$n_x = 195$	5.4	5.95	0.90
lens mesh, $\omega = 50$	$n_x = 260$	5.7	6.32	0.90
lens damping, $\omega = 40$	$\eta = 0.01$	135.1	5.80	23.32
lens damping, $\omega = 50$	$\eta = 0.01$	5.1	6.03	0.85
lens damping, $\omega = 40$	$\eta = 0.02$	78.0	5.82	13.41
lens damping, $\omega = 50$	$\eta = 0.02$	5.5	6.01	0.91
lens damping, $\omega = 40$	$\eta = 0.04$	37.9	5.77	6.57
lens damping, $\omega = 50$	$\eta = 0.04$	5.9	5.94	1.00
lens sponge, $\omega = 37.5$	width 0.6, $\lvert R\rvert = 0.31$	6.6	3.74	1.77
lens sponge, $\omega = 37.5$	width 1.2, $\lvert R\rvert = 0.12$	4.0	3.00	1.32
lens sponge, $\omega = 40$	width 0.6, $\lvert R\rvert = 0.34$	78.0	5.82	13.41
lens sponge, $\omega = 40$	width 1.2, $\lvert R\rvert = 0.11$	64.1	4.38	14.62
lens sponge, $\omega = 45$	width 0.6, $\lvert R\rvert = 0.33$	16.3	6.54	2.49
lens sponge, $\omega = 45$	width 1.2, $\lvert R\rvert = 0.11$	10.3	4.56	2.26
lens sponge, $\omega = 50$	width 0.6, $\lvert R\rvert = 0.32$	5.5	6.01	0.91
lens sponge, $\omega = 50$	width 1.2, $\lvert R\rvert = 0.11$	6.7	4.61	1.46

- **Absorbing layer.** The reflection coefficient is fitted on the centerline of the straight-fiber plate, $u_x = Ae^{-ikx} + Be^{ikx}$ with the complex along-fiber qP wavenumber, over $0.5 < x < 3.3$ (fit residual $\leq 2\%$); $|R| = |B/A|$ at the sponge entrance. The paper's sponge (width 0.6) gives $|R| = 0.31$–0.34 over $\omega \in [35, 50]$ (Table 11 lists 37.5–50); doubling its width outward, so that the interior is unchanged, gives $|R| \leq 0.13$. Table 11 lists the lens metrics for both.
- **Re-optimization with the better absorber.** The lens is re-optimized from straight fibers with the doubled sponge (same objective, MMA settings, 70 iterations and mirror-symmetric CS-RBF field; `lens_absorber.py`). At $\omega = 40$ it reaches $G = 70\times = 4.4\times$ (G_φ) $\times\, 15.9\times$ (growth), with $G_\varphi \geq 3.6\times$ over $[40, 60]$ and a band average of $3.9\times$, equal to that of the original design evaluated with the same absorber ($3.9\times$). The original design does not exploit the reflections; the reflections raise the redirection metric of any design.
- **Panel mesh.** The band design re-evaluated on a finer conforming mesh ($h = 0.04$ instead of 0.055; `panel_mesh_check.py`) keeps every qualitative result across the band: the sensor at or above $1.08\times$, the payload at or below $0.27\times$ and the rims at or below $0.53\times$ their straight-fiber energies. Sensor and payload ratios change by at most 6%; the rim ratios, the smallest energies, change most (by up to 0.10 in absolute terms).
- **Cloak frequency response on the finer mesh.** The cloak re-optimized on $h = 0.03$ gives $7.4\times$ and $14.6\times$ at $\omega = 47.5$ and 52.5 against $40\times$ at 50 (`cloak_fine_band.py`), as narrowband as the design on the coarser mesh.

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